Physics of Causality

Gravity as the response of a causal order: from a Lorentz-invariant substrate to the Einstein–Hilbert stiffness of the Benincasa–Dowker action

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Abstract

Causal Physics takes the causal order of events, not spacetime geometry, as the primitive of physics. We formulate it so that every step can be confronted with a pre-registered numerical test, and we report the outcome of these tests, positive and negative.

Substrate. The tests decide the nature of the substrate. A discrete substrate with a finite number of neighbours per event and a rest frame is not supported; the substrate is a Lorentz-invariant causal set, a Poisson sprinkling in which geometry is read from order and number. Matter is a flow of events that carries its own four-velocity, and the action of a flow is its proper time as counted by the order.

Gravity as the response of the order. Near matter the order changes: clocks slow down, causal cones narrow and events become denser. A fixed order gives the scalar (Nordström) class $\gamma = -1$, cones that only narrow give $\gamma = 1/d_s$, and an order that conserves its causal throughput gives $\gamma = 2/(d_s - 1)$, which is $\gamma = 1$ exactly in three spatial dimensions. In 3+1 dimensions this throughput principle is the linearised coupling of general relativity to the full energy–momentum tensor. A substrate rule in which every flow narrows the cone in its own rest frame and capacity deficits add reproduces this coupling on 3+1 sprinklings: two counter-streaming beams gravitate with the predicted active mass (2.40 ± 0.05 against 2.36, alternatives 1.56 and below), the cones become anisotropic and a single beam drags them.

The rule from the action of the order. With proper time counted by the order, the substrate rule is the linear response of an order whose action has the Einstein–Hilbert stiffness. We compute the expectation value of the smeared Benincasa–Dowker action of a causal set to second order in a static perturbation, in 3+1, by deterministic quadrature with nothing from general relativity put in. The Einstein–Hilbert stiffness emerges in the conformal sector, 6.0003 against 6, and in the mixed sector, where the perturbation changes the causal order itself, 7.983 against 8, and in the optical sector, where rays bend, 2.002 against 2. With all three coefficients $\gamma = 1$ follows from the action of the order (0.997 to 1.000). The discreteness corrects the stiffness non-analytically, $1 - \hat{K} \simeq (k\ell_k)^2[4.0 \ln(1/k\ell_k) + 0.9]$, and adds a non-local term that grows with the age T of the static field, $-0.10\,(k\ell_k)^3 T/\ell_k$, which we predicted analytically before computing it.

Predictions. Beyond the leading order the sectors differ, and each weakens with the age T of a static field: linearly in the conformal and mixed sectors, faster than linearly in the optical sector within the computed range. The age terms are not suppressed like the local corrections: if the age of a static field is cosmological, a non-locality scale at the level of existing bounds, $\ell_k \sim 10^{-19}\,\mathrm{m}$, places the breakdown of the static response at tens of micrometres or above, within reach of torsion balances. The asymptotic form of the optical age term, which fixes the far-field law of a point mass, is the next computation. Beyond the linear order the theory is completed by general relativity, to be derived thermodynamically; strong-field predictions of a scalar field equation are therefore not predictions of the theory. Every result quoted here is reproducible with the open Causality Lab [1], including the negative ones.

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1 Introduction

1.1 Causality before geometry

General relativity treats spacetime geometry as dynamical but primary; quantum field theory treats it as fixed and equally primary. In both, matter and fields evolve on a manifold that exists before anything happens on it. Causal Physics adopts the opposite stance: the primitive of physics is a set of events and the relation of influence between them,

$$u \prec v \quad \Longleftrightarrow \quad u \text{ can influence } v, \tag{1.1}$$

and geometry is a large-scale description of regularities in that relation.

This stance is not new. It is the founding hypothesis of causal set theory, compressed by Sorkin into the slogan "order + number = geometry" [2, 3]: a causal order determines a Lorentzian metric up to a conformal factor, and the number of events in a region fixes that factor. What Causal Physics adds is a physical reading of the order. Events are updates of a substrate; persistent flows of events are matter; and gravity is the way the order itself responds to those flows, slowing clocks, narrowing causal cones and adding events near matter. The question this paper asks is how much of gravitation follows from that picture once it is made precise enough to be computed.

1.2 Method: a laboratory that tests and does not rescue

All quantitative statements in this paper come from the Causality Lab [1], an open collection of numerical tests written for this programme. The rules of the laboratory matter as much as its results, because a theory developed together with its own tests can easily test only what it already put in.

Every test states its question, prediction, null model and numerical criterion before the full run, declares the level of conclusion it can support, and ends with one of four verdicts: PASSED, FAILED, UNDECIDED or INSTRUMENT FAILED. Negative results are kept, and a failed test is never rescued by changing its criteria. The rules are stated in full as the methodological principles M1–M7 (Sec. 29).

Throughout the paper, results are labelled with the identifiers of the tests (e.g. F7.4a) so that each can be rerun. Appendix A lists the tests used.

1.3 Main results

1. **The substrate** is a Lorentz-invariant causal set; a finite-degree substrate with a rest frame is excluded by the tests together with a known theorem (Sec. 18).
2. **Geometry is read blindly from the order.** An instrument that receives only events and their relation recovers the clock potential ϕ and the spatial potential ψ of a weak field, and with them the post-Newtonian parameter γ, in 2+1 and 3+1 dimensions (Sec. 19).
3. **Matter carries momentum.** A flow that keeps its timelike direction as a state stays localised and moves with its source, and its energy–momentum tensor is measured correctly (Sec. 20).
4. **The response of the order fixes γ.** A fixed order gives $\gamma = -1$, narrowing cones alone give $\gamma = 1/d_s$, and conservation of causal throughput gives $\gamma = 2/(d_s - 1)$; on 3+1 sprinklings the last reads $\gamma = 1.000 \pm 0.045$ (raw value; the formal verdict is undecided because the comparison metric was read with a larger error, Sec. 22).
5. **The coupling is tensorial.** In 3+1 the throughput principle is the linearised coupling of general relativity to T_{ab}; a substrate rule realising it makes pressure and motion gravitate and drags the cones (Part V).

6. **The rule follows from the action of the order.** The expected Benincasa–Dowker action of a causal set carries the Einstein–Hilbert stiffness in the static conformal sector, 6.0003 against 6, in the mixed sector, 7.983 against 8, and in the optical sector, 2.002 against 2; $\gamma = 1$ follows (0.997 to 1.000). The discreteness adds a non-analytic correction and non-local age terms (Part V).

7. **The road to predictions** is the static quadratic form of the order, now known in all three sectors; the asymptotic age term of the optical sector is the next computation (Part VII).

8. **Where to look.** The age term is not suppressed like the local corrections. For a cosmological age and ℓ_k at the level of existing bounds it places the breakdown of the static response at tens of micrometres, the range of current torsion balances (Sec. 45).

1.4 Relation to existing frameworks

Causal set theory. The substrate, the reading of geometry from order and number and the Benincasa–Dowker operator and action are taken from causal set theory [2–6]. The contribution of this work lies elsewhere: in the physical reading of matter as flows that carry momentum, in the principle by which the order responds to them, in the explicit second-order computation of the expected action with its non-local age term, and in the systematic numerical testing of each step.

General relativity. The theory does not compete with general relativity in the regime where general relativity is tested. In the linear regime it reproduces it, and beyond the linear order it adopts it, with the intention of deriving it thermodynamically along the lines of Jacobson [7]. Its own content lies in the discreteness and non-locality of the substrate.

Uniqueness theorems. A massless spin-2 field coupled consistently to the total conserved energy–momentum is general relativity [8, 9], and in four dimensions the Einstein tensor is the unique divergence-free tensor built from the metric and its first two derivatives [10]. The results of Part V are consistent with these theorems and use them: once the order is shown to couple to T_{ab} with the Einstein–Hilbert stiffness, the non-linear completion is fixed.

Modified gravity. The theory makes no claim of a modified low-acceleration regime (Sec. 42).

1.5 Structure of the paper

Part I describes, without equations, how Causal Physics sees the world: events, order and time, observation, transport, merging, matter, gravity, horizons and superposition, each statement labelled by how firmly it stands. Part II sets up the substrate, the reading of geometry from it, matter as flows and the response of the order. Part III states the parameters and the conditional bridge to SI units. Part IV collects the axioms in their revised form, with the status of each. Part V derives and tests the gravitational coupling, from the throughput principle to the Einstein–Hilbert stiffness of the discrete action. Part VI states what remains of the strong-field and astrophysical claims. Part VII sets out the predictions the theory can make and how they can fail. Part VIII discusses the limitations. Appendix A describes the laboratory and the tests; Appendix B relates this formulation to the first published version of the programme.

Part I
The Causal Picture of the World

This part describes the world as Causal Physics sees it, in words rather than equations. It follows one line of thought: from what exists, through time, space and motion, to matter, gravity and the limits of what can be known. The mathematics of the following parts is a precise form of this picture. Each statement carries a label that says how firmly it stands: [tested] – supported by a test of the laboratory (identifier given); [derived] – follows from a derivation in this paper or from established results it uses; [observed] – seen in a test, but not the outcome the test judged; [interpretation] – a reading consistent with the results, from which nothing is computed; [open] – part of the programme, not yet formulated or tested.

2 What exists: events and influence

The world is not made of things that exist in space and persist in time. It is made of *events*: elementary happenings, each of which is an act of realisation. An event has no size, no position, no mass and no charge. What it has is relations: some events can influence it, and it can influence others. Everything we call a property, the position of a stone, the mass of a planet, the tick of a clock, is a pattern in these relations, visible only when many events are taken together. [interpretation]

The single primitive relation is *influence*: one event can affect another. Influence has a direction, and it never closes into a loop: nothing influences its own past. With these two facts the set of all events becomes an *order*, and the order is the deepest structure of the world. Everything else in this part is a way of reading it. [derived]

A table, in this picture, is not an object that happens to be at a place. It is a pattern of events that keeps reproducing itself, update after update. Its solidity is the stability of the pattern, and its place is the way the pattern is woven into the surrounding web of influence.[interpretation]

3 Time: the order seen from inside

"Earlier" means "could have influenced"; "later" means "could have been influenced by". Time is nothing more than this. There is no universal now: two events that cannot influence each other have no fact about which came first, and no observer can find one. [derived]

Each observer has a local time, the number of events along their own history. A clock is a pattern that repeats, and its ticks are counted events. The clocks of two observers agree only where their histories are related, and they need not agree on how much time has passed between two meetings. [derived]

The passage of time is the growth of the order. New events appear in the future of existing ones and never in their past. The past is fixed because it has already been realised; the future is open because it has not. [interpretation]

4 Space and dimension

Space is what lies between events that cannot influence each other. A "moment" of the world, as an observer experiences it, is a large set of mutually unrelated events, and the distances within it are read from how far one has to go into the past or the future to find events related to both. Space is therefore not a container that exists first; it is the side-by-side structure of the order. [derived]

The order fixes the shape of spacetime up to a local scale, and counting events fixes the scale. Nothing else is needed: distances, volumes, the rate of clocks and the dimension itself are read from order and number. The dimension, for example, is visible in what fraction of pairs of events in a region are related. [tested: K.1, K.16b, K.16c]

Why space has three dimensions is not derived here. But it is notable that the way the order responds to matter (Sec. 11) agrees with the gravity we observe in exactly three spatial dimensions and in no other. [derived] Whether this is the reason for three dimensions is an open question. [open]

5 Motion and relativity

The substrate looks the same to every observer moving uniformly. There is no state of rest in it, no ether against which motion could be measured, and no experiment can find one. This is the principle of relativity, and here it is a property of the substrate, not an additional postulate. [derived]

The speed of influence is the same for all observers for the same reason: the boundary between events that can and cannot influence each other is part of the order, and the order is the same for everyone. The speed of light is the conversion between counting events along time and counting them across space. [derived]

A moving clock runs slow because it takes a shorter path through the order. Between two meetings, the observer who stays at rest passes through the longest chain of events, and an observer who travels away and comes back passes through fewer. The travelling twin is younger because fewer events happened to her, not because anything acted on her clock. [derived]

6 Observing is receiving causes

To observe something is to be on the receiving end of its influence. Seeing a tree means that events in the tree have influenced, through a long chain of intermediate events, the events of our eye. There is no observation without causal contact, and every observation is itself an event that joins the order.

This explains why the world looks different near and far. A tree next to us is rich in detail; the same tree on the horizon is a silhouette. The tree has not changed. What has changed is the density and fidelity of the influence that connects it to us. Two different effects are at work.

- *Geometric dilution.* The same influence spreads over a larger sphere as it travels, so less of it reaches us, in proportion to $1/r^2$. This is the root of the inverse-square law.[tested: F1.1]
- *Loss through intermediaries.* Where influence passes through many intermediate events (air, dust, memory, a historical record), each of them reshuffles it, and detail is lost. This is the entropic side of distance. [interpretation]

Influence that travels through empty space along the light cone does not pass through intermediaries in this sense: events on the light cone are causally close however distant they are in space. The images of very distant point sources are therefore not blurred by the discreteness of the substrate, and the sharp images of distant quasars agree with that. [derived]

7 Scale and causal resolution

What we call scale is the resolution with which the order can be read. Different physical regimes are different limits of that resolution, not different sizes.

The discreteness scale. At the scale of individual events the order is well defined, but there are too few events in any region to fix a shape. Trajectories, fields and distances lose their meaning here, and only the relations remain. [derived]

The plateau. When regions contain many events and the order is close to a uniform sprinkling, geometry becomes an excellent compression of it. This is the regime of ordinary matter and of everyday experience, in which three-dimensional space and Lorentzian spacetime are reliable descriptions. Dynamics of orders of this kind has a phase whose typical members look like spacetime, separated by a transition from phases that do not. [known in 2D [11]; in part F6.4a] General relativity and quantum field theory are effective descriptions on the plateau.[interpretation]

Between the two. At an intermediate scale ℓ_k the order is smooth enough for geometry but still remembers that it is discrete. Its response reaches along the light cone to distant events, and the response to a static field depends on how long that field has existed. [derived, Sec. 35]

8 Transport: nothing from nothing

The substrate is not a passive stage. It carries influence from events to the events they affect, and this influence is conserved: what flows into an event flows out of it, apart from what the event stores. Nothing appears from nothing and nothing disappears into nothing. This continuity is the most basic law of the substrate, and the conservation laws of physics are its large-scale expressions. A steady source surrounded by any closed surface sends exactly its own output through it, which is Gauss's law. [tested: F1.1]

The capacity of an event to pass influence on is finite. When more influence arrives than an event can carry, it is held back: the substrate buffers it, and an observer sees this as delay. Delay is not a defect of transmission but the visible expression of a finite causal bandwidth.[interpretation]

9 Branching, merging and the meaning of energy

Influence branches and merges. *Branching* distributes the influence of one event among many later events. *Merging* compresses influence that arrived along distinct paths into a single continuation. At the level of the substrate the two are symmetric: every event is a meeting point of its past and a source of its future. [interpretation]

Energy. The amount of influence carried is *energy*; the direction in which it is carried is *momentum*. Together they are one quantity seen from different frames: energy is the part along time, momentum the part across space. Both are conserved because influence is. [tested: K.15]

Mass. When two streams of influence moving in different directions merge, their directed motion is not lost. It becomes internal: the merged stream moves more slowly as a whole and carries more rest energy, in exactly the proportion that conserves energy and momentum together. Merging is how motion becomes mass and heat; two counter-streaming populations that merge cool down. Mass is energy that no longer points anywhere. This is the content of $E = mc^2$ in this picture. [tested: K.15, F4.1]

10 Light and matter

Light. Influence that travels on the boundary of the order, along the light cone, has no clock: no events happen along its way, it cannot be at rest, and it has no rest frame. Light is influence in pure transit. [derived]

Matter. Matter is influence that persists. A flow that only spread out would be radiation: it would fill its light cone and dissolve. A flow that stays together, carried from event to event along a narrow bundle of histories, is matter, and along its history events happen: matter has clocks.

What keeps a flow together on a substrate that has no preferred directions is *memory*. A flow that carries its own direction of motion as a state passed from event to event stays on the world line of its source and moves with the source when the source moves. This memory of direction is *inertia*: a body continues in its motion because its flow remembers where it was going. Momentum is not added to matter from outside; it is what makes localised matter possible at all. [tested: F3.6]

Structure. Where repeated activity keeps routing influence through the same region, the substrate approaches its capacity and influence can no longer spread freely: it is confined to a restricted set of paths. Structure is the self-maintaining outcome of this confinement, a dynamical equilibrium between the continuity of influence and the finiteness of capacity, maintained as long as the activity that creates it continues. [interpretation] How such confinement closes into a self-bound object, a particle that holds itself together without an external source, is the central open problem of the matter sector. [open]

11 Gravity: the order responds to matter

Load. Matter is a flow that passes through the substrate continuously, and the substrate carries it as load. A loaded substrate does three things at once: its clocks run slower, because more of its activity goes into carrying the flow; its causal cones narrow, because influence travels more slowly through it; and it becomes denser, because it adds events.[effects derived; reasons: interpretation]

Throughput. The last two effects are tied together by one principle, familiar from traffic and from every conserved current passing through a constriction: *the throughput is conserved.* Where cars slow down they bunch up, so that the number passing a point per minute stays the same. Where the substrate slows influence down, it adds events, so that the number of events a light front passes per unit time and area stays the same. The speed of light is the maximal rate at which influence travels without congestion; near matter the effective speed falls below it. [derived; tested: F7.1c]

The field. The response of the order is not confined to where the matter is. It spreads, at the speed of influence, into the surrounding substrate, weakening with distance as influence does. The gravitational field is this spreading response of the order: not a substance added to spacetime, but the order itself, reshaped. [tested: F1.3]

Falling. A body falls because its history bends towards slower clocks: of all histories between two events, matter follows the one along which the most events happen, and near a mass that history leans towards the mass. Light bends because the cones it travels along narrow towards matter. The amount by which light bends, twice what a naive Newtonian picture gives, is exactly what the conservation of throughput gives in three spatial dimensions. [derived; fall from the order: 2+1 passed (F5.1d), 3+1 failed (F5.1e)]

Equivalence. All bodies fall in the same way, whatever they are made of, because falling is not a reaction to a force but the following of the order's own longest histories. What a body is made of does not enter into which history is longest. Being pushed and being attracted probe the same property of the substrate from two sides. [derived]

Everything that moves gravitates. The substrate responds to a flow in the flow's own rest frame, and the responses of different flows add. Because of this, not only mass but also pressure and motion gravitate, and a moving flow drags the causal cones along with it. [tested: F7.4a] The order responds in this way because its own dynamics, the action that counts its intervals, has exactly the stiffness of Einstein's gravity. [derived and computed, Sec. 35]

Non-linearity. Weak fields add. Strong ones do not: when two sources sit close together, each lies in the well of the other and acts more weakly, so the common well is shallower than the sum of the two. Gravity is non-linear because the order responds to its own response. [tested: F1.4]

12 Horizons and the limits of reconstructability

Where the substrate is loaded very strongly, geometry stops being a good description of the order. The events are still there and influence still flows, but the order no longer behaves like a smooth spacetime: reading it with clocks, cones and densities gives answers that do not fit together into one metric. [observed: F7.1b]

A *causal horizon* is the limit of reconstructability: the boundary beyond which an external observer can no longer keep an order-preserving description of what happens inside using a global time, a smooth geometry or isotropic propagation. Beyond it, causal updates continue without interruption and causal continuity is intact. What is lost is the ability to describe the inside in the language of the outside. Horizons mark the reach of geometric language; they are not a boundary of causal evolution itself. [interpretation]

Merging plays a central role here. When many causal channels converge under extreme load, the causal distance between events collapses from the substrate's own point of view. To an observer who relies on geometric reconstruction, this collapse appears as temporal ambiguity, loss of detail and horizon behaviour. [interpretation]

13 Entropy and the arrow of time

The order has a direction because it grows: events are added to its future, never to its past. This is the deepest arrow of time. [interpretation]

Merging gives the arrow a second, familiar face. Branching can be followed backwards: every branch remembers where it came from. Merging cannot be undone uniquely: once distinct paths have joined into one continuation, many different pasts are compatible with the same present. What is lost is the record of which past it was, and the measure of that loss is *entropy*. Irreversibility is the asymmetry between how easy it is to merge and how hard it is to unmerge. [interpretation]

A horizon carries an entropy proportional to its area, and the order counts it directly: the number of pairs of events that straddle the horizon in a minimal way is proportional to its proper area. [tested: K.14] This is the bridge to the thermodynamic view of gravity, in which Einstein's equations describe the equilibrium of the order the way the gas laws describe the equilibrium of molecules. [open, Sec. 38]

14 The vacuum is not empty

Empty space is not nothing. It is the substrate with no flow through it: events at their fundamental density, related to each other in the way that makes a flat spacetime. Every region of vacuum contains a definite number of events, and that number fluctuates as counts do, by about its square root. [derived]

Such fluctuations of number act on the large-scale geometry like a small cosmological constant whose size is set by the volume of the observable universe; for the present universe it is of the order of the observed value [12]. In Causal Physics this would make dark energy the residue of the discreteness of the substrate. [open]

The growth of the order on the largest scale is the natural candidate for the expansion of the universe. [open]

15 Superposition and measurement

An event can have several admissible continuations, and as long as nothing distinguishes them they coexist. In Causal Physics there is no separate collapse process; there is stabilisation under interaction.

- *Superposition* is the coexistence of admissible causal continuations.
- *Measurement* is a causal interaction that biases the routing of influence towards one of them.
- *Collapse* is the stabilisation of a single continuation. It is irreversible because undoing it would require a coordinated redistribution of influence across an extended region of the substrate, and the growing impedance of that region suppresses it. It is a merging that cannot be undone.

At the discreteness scale, where patterns do not persist long enough to be compressed into trajectories, every observation perturbs what it probes. Quantum behaviour is then the behaviour of a substrate that is too unstable for geometric compression, rather than a paradox of measurement. [interpretation] A quantum dynamics of the order that would make this picture quantitative is not part of the present work. [open]

16 The frontier

Persistent currents of influence that circulate rather than stream are natural candidates for *charge* and *spin*: asymmetries in how influence is routed around a structure, invisible from far away except through their effect on routing. The freedom to relabel routes without changing any observable flow would then be the origin of *gauge freedom*. These ideas belong to the programme; none of them is formulated or tested here. [open]

17 The picture in one paragraph

The world is a growing order of events. Its direction is time, its side-by-side structure is space, and its sameness for every observer is relativity. Influence flows through it and is conserved; its amount is energy and its direction is momentum. Influence that persists and remembers its direction is matter, and that memory is inertia; influence in pure transit is light. Matter loads the order, and the order responds by slowing its clocks, narrowing its cones and adding events so that its throughput is kept: that response, spreading outwards, is gravity. Where the load is extreme, geometry can no longer describe the order, and a horizon marks the limit of what the outside can reconstruct. What cannot be unmerged is entropy, and what has not yet been stabilised is superposition. From events to gravity each step is tested or derived; self-bound matter, charge, spin and the quantum are the frontier.

Part II
Causal Physics on a Lorentz-Invariant Substrate

18 The substrate

18.1 Events, order and number

The primitive objects are *events*. An event carries no position, duration, mass or charge; everything is relational. Events are connected by a relation of influence $\prec$ that is

- *transitive*: $u \prec v$ and $v \prec w$ imply $u \prec w$;
- *acyclic*: $u \prec v$ and $v \prec u$ never hold together (no closed causal loops);
- *locally finite*: between any two events there are finitely many others.

A set with such a relation is a causal set [2, 3]. Two derived notions carry most of the physics:

- the *interval* $I(u, v) = \{w : u \prec w \prec v\}$ and its number of elements $n(u, v)$;
- the *chains* $u \prec w_1 \prec \cdots \prec v$; the length of the longest chain between u and v is a discrete proper time.

When the causal set is faithfully embedded in a Lorentzian manifold (M, g), the relation $\prec$ is the causal order of g and the number of elements in a region is, on average, ρ times its spacetime volume. The order fixes the metric up to a local scale; the number fixes the scale. The standard realisation is a *sprinkling*: a Poisson process of density ρ per unit proper volume, with the relation inherited from the light cones of g. A sprinkling is statistically invariant under Lorentz transformations: there is no preferred frame in which it looks different from any other.

18.2 Why the substrate must be Lorentz invariant

A natural alternative is a graph in which each event has at most K_0 relations, locality is distance in the graph and time is the count of updates. Such a substrate has a rest frame. A Lorentz-invariant discrete structure cannot have it: in a sprinkling every event is related to infinitely many others along its light cone (finitely many only because a simulated region has a wall), and no Lorentz-invariant rule can select a finite set of neighbours locally [13]. Finite degree and Lorentz invariance of the substrate exclude each other.

There are two options. (A) A Lorentz-invariant substrate: relativity comes for free and non-locality is unavoidable. (B) A substrate with a rest frame: Lorentz invariance must emerge for observers with extreme precision, and small violations at the discreteness scale tend to be amplified by quantum corrections. The tests of the laboratory support (A) (Appendix A):

- the degree of an event grows with the number of events in the region; a bound K_0 does not hold (F0.3);
- capacity limits the *flow* through an event, not the number of its relations (F1.8; supported, but not robust to the seed);
- local growth rules of the finite-degree type produced no manifold-like order (0 of 7 rules, F6.1) and no three-dimensional space (F6.2).

The substrate of Causal Physics is therefore a causal set. Capacity survives as a bound on the flow that an event can carry (Part IV).

18.3 Non-locality

Because every event is related to events arbitrarily far away along its light cone, local differential operators have no direct discrete analogue. The causal set d'Alembertian of Benincasa and Dowker [4], in its smeared form due to Sorkin [5], sums over the whole causal past of an event with weights that depend on the number of elements in each interval. The smearing introduces a non-locality scale

$$\ell_k = (\epsilon\rho)^{-1/D}, \tag{18.1}$$

where D is the spacetime dimension and $0 < \epsilon \leq 1$ the smearing parameter. For $\epsilon = 1$, ℓ_k is the discreteness scale itself; for small ϵ it is larger. The operator is defined in Sec. 24; its non-locality is a physical property of the substrate, and one of the results of this paper (Part V) is a non-local term in the action that it implies.

19 Reading geometry from the order

A theory in which geometry is emergent must say how geometry is read. For a weak static field in isotropic coordinates,

$$\mathrm{d}s^2 = -e^{2\phi}\,\mathrm{d}t^2 + e^{-2\psi}\,\mathrm{d}\boldsymbol{x}^2, \qquad \gamma = \psi/\phi, \tag{19.1}$$

(ϕ the clock potential, ψ the spatial potential, γ the post-Newtonian parameter), three quantities of the order determine both potentials:

quantity of the order	what it measures	weak-field value
clocks: elements of a small interval at rest	proper time	ϕ
the boundary of the relation	speed of the causal cone c	$\phi + \psi$
density of elements per coordinate volume	$\sqrt{-g}$	$\phi - d_s\psi$

(d_s is the number of spatial dimensions.) Any two of the three determine γ; the third is a consistency check that the order is metric at all.

The blind instrument. The laboratory built an instrument that receives only the events, the relation and the fundamental density, not the metric, and reads γ along two independent paths: (A) cones and density, (B) clocks and cones. It was calibrated on orders sprinkled into metrics with known γ (K.16b in 2+1, K.16c in 3+1). In 2+1, for a field $\phi = -0.1\,e^{-r^2/128}$, it returns -1.000 ± 0.001 for $\gamma = -1$, -0.04 ± 0.02 for $\gamma = 0$ and $+0.92\pm0.03$ for $\gamma = +1$. In 3+1 it returns -1.001 ± 0.001, 0.33 ± 0.04 and 0.99 ± 0.05 for $\gamma = -1,\ 1/3,\ +1$. For fields weaker than $|\phi| \sim 0.03$ the instrument is unreliable; all tests that use it work with $|\phi| \sim 0.1$. The instrument uses coordinates and the known density as a declared readout; a coordinate-free reading (radar echoes) is a planned improvement. Figure 1 shows the calibration and the potentials read by radius.

Other instruments. The clock rate Π can be read back from the density of chains (K.3); the causal front of a source measures the Shapiro delay to within 2% of the general relativistic value (K.2); and the number of "horizon molecules" [14, 15] measures the area of a horizon in flat space in 1+1, 2+1 and 3+1 dimensions and its *proper* area in a conformally flat geometry, with the exponent of the proper-area law 1.96 ± 0.06 against 2 in 3+1 (K.14). The last instrument is the entropy meter of the thermodynamic route of Sec. 38.

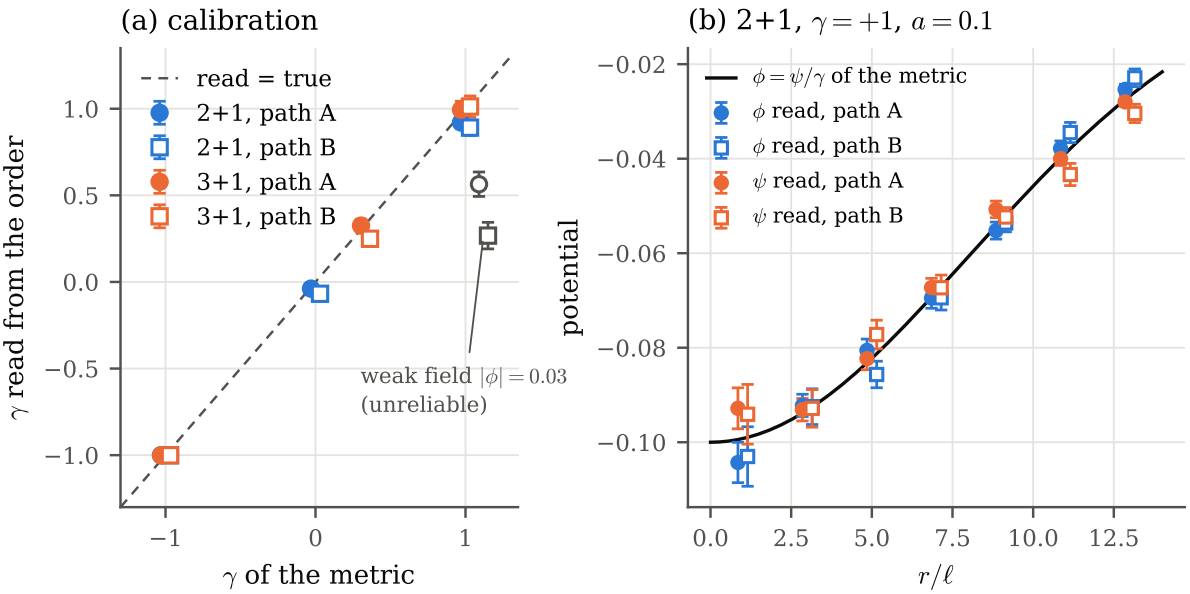


Figure 1: The blind instrument (K.16b, 2+1; K.16c, 3+1). (a) γ read from orders sprinkled into metrics of known γ, along path A (cones and density, filled) and path B (clocks and cones, open); grey: the weak field $|\phi| = 0.03$, where the instrument is unreliable. (b) The potentials read by radius for $\gamma = +1$ in 2+1 ($a = 0.1$, $\psi = \phi$) against the profile of the metric. Bars: ± 1 standard error over 32 (2+1) and 16 (3+1) sprinklings.

20 Matter as flows

20.1 Flow and its conservation

Matter in Causal Physics is a persistent pattern of events. Its simplest model is a *flow*: a non-negative quantity F transported from events to their successors along the relations of the order and conserved at every event (Axiom 4). A steady source then obeys Gauss's law: the flow through any closed surface around it equals the source, $J/S = 1.003$–1.004 in 2+1 and 3+1 with up to 2.6×10^5 events, against 0.77–0.88 for a model without conservation (F1.1).

20.2 A flow carries its four-velocity

How the flow is shared among successors decides whether anything localised can exist. If an event divides its flow equally among all successors, a source spreads at $0.77c$. If each step takes its direction from the previous step, the memory decays within a few relations (a single relation turns the direction by 0.45–0.52 in rapidity) and the flow spreads at $\sim 0.4c$. If the flow carries its timelike direction as an internal *state*, the bundle stays at the world line of its source, $v_s = 0.054 \pm 0.006\,c$ (F3.6, replicated). A source moving at $0.5c$ is followed: the bundle is at rest in the frame of the source ($+0.008 \pm 0.004$ against the alternative -0.5), and its width evolves as for a source at rest, as the principle of relativity requires. Recomputing the rule in a frame boosted by $0.6c$ gives the same choice at every event.

This was the first result of the laboratory that had not been put into the rule. It shows that localisation on a Lorentz-invariant substrate requires a memory of direction, and that the memory is inertia: a flow carries momentum.

20.3 The energy–momentum tensor of flows

With a four-velocity u^a per carrier, the flows have an energy–momentum tensor

$$T^{ab} = \frac{1}{\Delta V_4} \sum F\, u^a u^b\, \tau, \tag{20.1}$$

summed over the relations in a spacetime cell of volume ΔV_4, with τ the proper duration of each relation. The laboratory measures it correctly for dust, for two counter-streaming populations and for an isotropic gas (K.15): for a beam at $v = 0.6$ the cell values are $E/e_0 = \gamma_v^2 = 1.5625$ (e_0 the rest-frame energy density), $|q|/E = v$, $S_{xx}/E = v^2$ without scatter, and the pressure of an isotropic gas is reproduced to within 7%. When flows of different directions merge, the rule $F = |P|$ (the merged flow is the length of the summed four-momentum) conserves four-momentum exactly (6×10^{-17}), while the earlier rule "flow = sum of flows" loses up to 2% of the energy. Conservation $\partial_a T^{ab} = 0$ is the condition without which a tensor coupling to gravity is inconsistent (Part V).

20.4 Contact: flows that meet stick

Every event passes its whole flow to one successor, so two flows interact by contact only if their chains choose the same events; a priori they might cross each other without interacting. They do not. Two flows sent head-on at ± 0.6 merge on contact: after the crossing 77–82% of the flow is on events that carry both (0.821 ± 0.013 in 2+1, 0.774 ± 0.008 in 3+1), the merged flow is at rest in the centre-of-momentum frame (-0.008 ± 0.013 and -0.003 ± 0.011), its rest mass exceeds the sum of the merged rest masses by up to the Lorentz factor (1.216 against $\gamma = 1.25$ for equal parts), and no flow reverses its motion. Flows displaced so that they do not touch do not merge (F4.1, PASSED in both dimensions). Contact between flows is a perfectly inelastic collision: momentum is shared, kinetic energy becomes rest mass, and the inertia of the merged flow is its energy. The rule has no elastic channel; elastic scattering of matter must come from fields or from bound structure, which the model does not yet have.

21 The action of a flow

The Causal Resistance Functional $\mathcal{R} = \sum_v Z_v \Delta\tau$ (impedance Z_v times the causal update increment) is the action of a flow along its history. Its content depends on what $\Delta\tau$ is. Here $\Delta\tau$ is the proper time between successive events of the flow, as counted by the order (the length of the longest chain, or equivalently the number of elements of a small interval). Then, for a flow with mass m,

$$S_m = -\sum m\, \Delta\tau \;\longrightarrow\; -m \int \mathrm{d}\tau, \tag{21.1}$$

the action of a free particle (Axiom 8).

Two consequences follow without further assumptions.

1. **On a fixed order the geometry of motion is conformal.** If clocks are slowed by a factor Π but the order is the flat one, the action is $\int \Pi \sqrt{\mathrm{d}t^2 - \mathrm{d}\boldsymbol{x}^2}$, the geodesic action of the conformally flat metric $\Pi^2\eta$. Maximal chains on sprinklings follow exactly these geodesics (F2.2, F2.4, F2.5 in 3+1). But null geodesics of a conformally flat metric are straight: light is not deflected, $\gamma = -1$.

2. **The coupling to the order is through T_{ab}.** For a small change of the geometry $g = \eta + h$ the proper time changes by $\delta\tau = -\frac{1}{2} h_{ab} u^a u^b \tau$, so

$$\frac{\delta S_m}{\delta h_{ab}} = \tfrac{1}{2}\, T^{ab}. \tag{21.2}$$

The order "sees" a flow only through $u^a u^b$, that is in the rest frame of the flow, and contributions of different flows add. These two facts are the substrate rule of Sec. 33, derived rather than assumed.

22 How the order responds to matter

If the order near matter were fixed, gravity would be the scalar Nordström class and light would not bend. Gravity in Causal Physics therefore *changes the causal order*: near matter causal cones narrow and events become denser. Causality itself is untouched: there are no closed loops and the flow is conserved. How much the cones narrow relative to how much the density grows is a single number, and it is γ:

response of the order	cones	density	γ
fixed order	unchanged	unchanged	-1
cones narrow, events kept (capacity)	narrower	unchanged	$1/d_s$
causal throughput conserved	narrower	denser, $\rho_c\, c = \text{const}$	$2/(d_s-1)$
general relativity (3+1)	$c = e^{2\phi}$	$\rho_c = e^{-2\phi}$	1

The throughput principle. Where the substrate slows the propagation of influence, events become denser in such a way that the number of events crossed by a light front per unit time and area stays the same,

$$q = \rho_c\, c = \text{const}. \tag{22.1}$$

This is the continuity law of traffic (flow = density of cars × their speed) and of any conserved current through a constriction. It makes precise the picture of c as the maximal congestion-free throughput of the substrate, with an effective speed that falls where the substrate is loaded. With the dictionary of Sec. 19, $\ln(\rho_c c) = 2\phi - (d_s-1)\psi$, so

$$\gamma = \frac{2}{d_s - 1}, \tag{22.2}$$

which is $\gamma = 1$ exactly in three spatial dimensions. In 3+1 the principle together with a clock rate Π fixes the metric completely, $\mathrm{d}s^2 = -\Pi^2 \mathrm{d}t^2 + \Pi^{-2}\mathrm{d}\boldsymbol{x}^2$, with $c_{\text{eff}} = \Pi^2$ and $\rho_c = \Pi^{-2}$.

Tests. The three responses were built as discrete orders and read with the blind instrument.

- *Cones only* (F7.1a, 2+1): $\gamma = 0.47 \pm 0.03$ and 0.43 ± 0.03 against $1/2$, independent of the strength of the coupling, as derived before the run: narrowing the cones without adding events cannot give $\gamma = 1$. In 3+1 the same rule reads 0.35 ± 0.02 against $1/3$.
- *Throughput* (F7.1b, 2+1): $\gamma \approx 1.85 \pm 0.1$ against the prediction 2, in two independent runs; formally UNDECIDED because the two reading paths deviate in opposite directions by 0.1–0.2.
- *Throughput* (F7.1c, 3+1): $\gamma = 1.000 \pm 0.045$ (path A) and 0.972 ± 0.074 (path B); replication 0.948 ± 0.047 and 0.979 ± 0.058. The raw readings agree with the prediction, and the orders are metric (the two paths agree). Formally the test is UNDECIDED: it compares with an exact $\gamma = 1$ metric read by the same instrument in the same run, and in path B that control read low (0.78 ± 0.08 and 0.90). The design left too little room for the statistical error of the comparison; the verdict stands as recorded.

All readings, with the controls, are shown in Fig. 2.

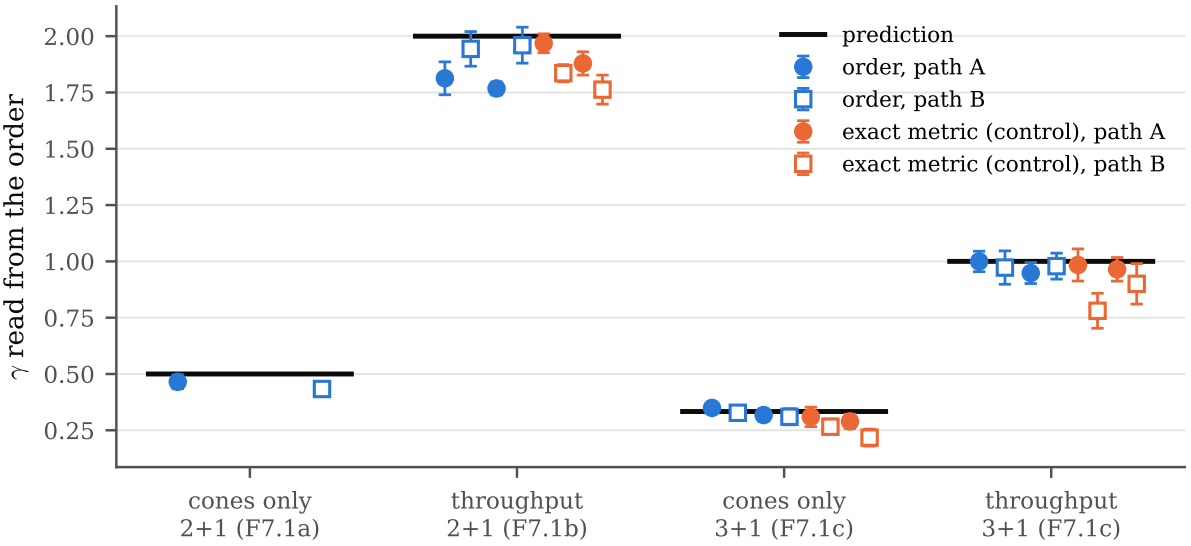


Figure 2: γ of the responses of the order (F7.1a; F7.1b and F7.1c, two independent runs each), path A (filled) and path B (open), against the prediction (black bar). Orange: orders sprinkled into the exact metric with the predicted γ and read by the same instrument in the same run (the controls of F7.1b and F7.1c); in 3+1 path B of the control reads low, which leaves F7.1c undecided. Bars: ± 1 standard error.

What this does and does not show. The tests show that a discrete order built according to the throughput principle has the geometry of $\gamma = 1$ in 3+1. They do not show why the substrate conserves its throughput. Part V answers that question at the linear order: the principle is the linearised coupling of general relativity, and it follows from an action of the order with the Einstein–Hilbert stiffness, which the expected Benincasa–Dowker action has.

23 Free fall in the order

The order of Sec. 22 carries gravity: its longest chains between two fixed events sag towards the mass. A separate question is whether matter, moved by a local rule that reads only the order, falls. Such a rule contains no bending law and no field; the fall has to come from inertia and from the order itself.

Set-up. The 2+1 throughput order of F7.1b with $\sigma_0 = 0.45$: density e^{σ}, cones $e^{-\sigma}$, $\sigma = \sigma_0 \exp(-r^2/2w^2)$, $w = 8$. With the dictionary of Sec. 19 its clock potential is $\phi = -\sigma/3$ ($\gamma = 2$), which predicts the Newtonian acceleration a_{pred} of a slow body. A history is continued as a longest chain: from its last event v, with the event p eight steps back, the rule looks at the events q eight links ahead, $L(v,q) = 8$, and steps along a longest chain towards the one with the smallest $L(p,q)$, i.e. the straightest continuation by the reverse triangle inequality. The reading is $R = a/a_{\mathrm{pred}}$, with a from a quadratic fit of the mean radial displacement over 22 units of time. Fixed before each run: the history falls as predicted if $\frac{1}{2} < R \mp 2\,\mathrm{SE} < 2$; in the flat order neither the histories may drift nor the longest chains sag, and in the loaded order the longest chains must read the geometry, $\frac{1}{2} < R_{\max} \mp 2\,\mathrm{SE} < 2$.

Results.

- A direction carried in flat coordinates (F5.1, carried) keeps its direction and does not fall, $R = -0.11 \pm 0.04$ (the control read $R_{\max} = 0.63 \pm 0.07$, narrowly below its band:

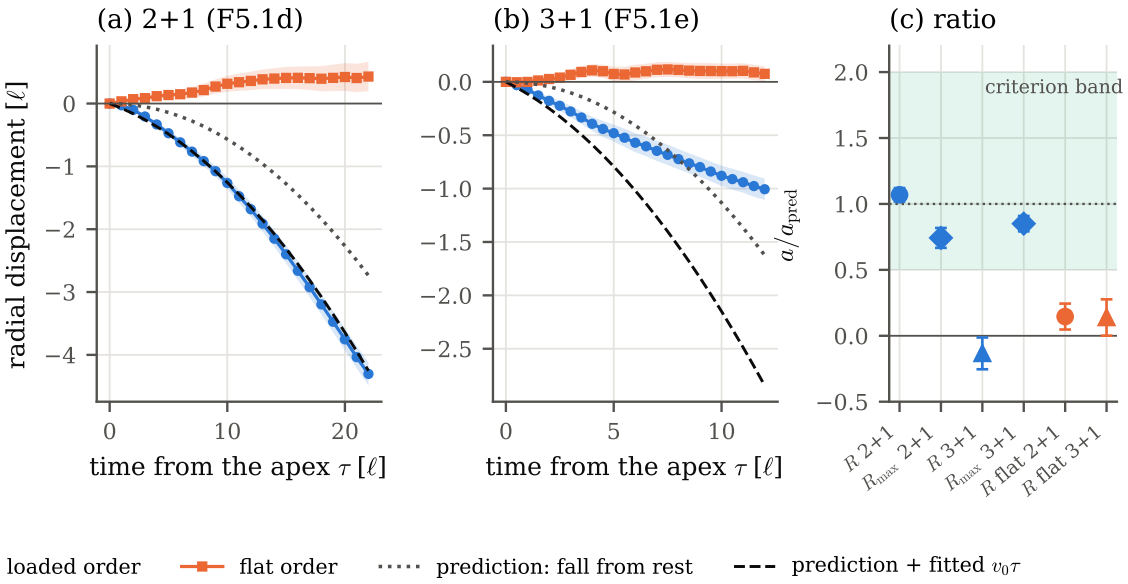


Figure 3: Free fall from the apex. (a) F5.1d, 2+1, 32 sprinklings per order, 40 histories each; (b) F5.1e, 3+1, 16 sprinklings per order, 60 histories each. The mean radial displacement from the apex (band ±1 standard error across sprinklings) in the loaded and the flat order; dotted: the Newtonian fall from rest at the apex radius, the prediction of the test; dashed: the same fall plus the fitted start velocity, $v_0\tau$. (c) $R = a/a_{\text{pred}}$ and the control $R_{\max}$ of both tests with the criterion band $\frac{1}{2} < R < 2$. Bars: ±1 standard error.

INSTRUMENT FAILED). A rule that keeps to the layer of the longest chains from the origin (F5.1, intrinsic) moves outwards, $R = -0.30 \pm 0.07$: FAILED.

- The longest-chain rule started from a virtual history at rest (F5.1c) moves inwards but not with the Newtonian acceleration, $R = 0.34 \pm 0.05$: FAILED. A history at rest is not a free state of the order.
- The same rule started at the apex of a thrown ball (F5.1d): the history is the rising half of the longest chain between two events at the same place, a free state read from the order. It falls, $R = 1.069 \pm 0.051$: PASSED. Flat order $R = 0.15 \pm 0.10$; longest chains $R_{\max} = 0.74 \pm 0.08$ (loaded) and 0.10 ± 0.12 (flat).
- The same experiment in 3+1 (F5.1e: the order of F7.1c, $\sigma_0 = 0.3$, $w = 4$, $\phi = -\sigma/2$; rule and criteria unchanged) does not fall, $R = -0.13 \pm 0.12$: FAILED. The instruments passed: flat $R = 0.14 \pm 0.14$, and the longest chains between fixed events read the geometry better than in 2+1, $R_{\max} = 0.85 \pm 0.06$. The history takes an inward velocity at the apex and then keeps it, without acceleration (Fig. 3b).

What this does and does not show. In a 2+1 order shaped by matter, a history that continues as a longest chain falls with the Newtonian acceleration, without a bending rule, provided it starts from a free state of the order. In 3+1 the same rule does not fall. So the coupling of matter to the gravity of the order is not yet solved: the order carries gravity in both dimensions (its longest chains between fixed events read it), but this rule passes it on to a moving history only in 2+1. In both dimensions the history receives an inward velocity jump at the apex (−0.09 in 2+1, −0.17 in 3+1, against +0.02 and −0.05 in the flat orders): the order acts on it at the start as a kick, and only in 2+1 does a continuous acceleration follow. A possible cause, not tested: in 3+1 the effect of the field on the chain lengths over the horizon of

the rule is below one link, so the choice of the next event is decided by the tie-breakers rather than by the geometry. Further limits of the 2+1 result: one run; the start from the apex was introduced after F5.1c failed, with its criteria fixed before the run; the fall is 1.44 times what the longest chains read of the geometry; the flat order drifts slightly outwards, within its tolerance. A replication of F5.1d and a rule that carries the acceleration in 3+1 are the next steps.

24 The Benincasa–Dowker operator and action

For a causal set sprinkled at density ρ into a D-dimensional spacetime, the smeared Benincasa–Dowker (BD) operator acting on a field f is [4–6]

$$(B_\epsilon f)(x) = -a_D(\epsilon\rho)^{2/D}\Big[f(x) + \sum_{y\prec x} K_\epsilon(n_{xy})\, f(y)\Big], \tag{24.1}$$

$$K_\epsilon(n) = \epsilon(1-\epsilon)^n \sum_i C_i \binom{n}{i}\Big(\frac{\epsilon}{1-\epsilon}\Big)^i, \tag{24.2}$$

where n_{xy} is the number of elements strictly between y and x. In 3+1 the coefficients are $C = (-1, 9, -16, 8)$ and $a_4 = 4/\sqrt{6}$. Its expectation over sprinklings is

$$\mathbb{E}[B_\epsilon f](x) = -a_D(\epsilon\rho)^{2/D}\Big[f(x) + \epsilon\rho\int_{J^-(x)} \sqrt{-g(y)}\; g_k\big(\epsilon\rho V(x,y)\big) f(y)\, \mathrm{d}^D y\Big], \tag{24.3}$$

with $V(x,y)$ the proper volume of the interval and

$$g_k(w) = e^{-w}\sum_i C_i \frac{w^i}{i!}. \tag{24.4}$$

In the continuum limit it tends to $\Box_g - \frac{1}{2}R$ in curved spacetime [4, 16]. The laboratory confirmed this limit with its own instruments, for conformally flat geometries (K.11, K.11c) and, separately for the density part and the cone part, for the geometry with bent light (K.13a: within 0.8–2.4% at a field width of $32\ell_k$).

The BD *action* of a causal set is

$$S_{\mathrm{BD}} = -\frac{2}{\rho}\sum_x (B_\epsilon 1)(x), \tag{24.5}$$

whose expectation tends, in the continuum, to the Einstein–Hilbert action $\int \sqrt{-g}\, R$ plus boundary terms [17]. In this paper the BD action is the candidate dynamics of the order. Part V computes its expectation to second order in static perturbations and compares the result with the Einstein–Hilbert value, which enters only as the normalisation of the comparison.

Part III
Units, Parameters and the SI Bridge

Causal Physics is formulated at the level of causal structure, not of spacetime geometry. Its fundamental quantities are therefore dimensionless and relational: counts of events, relations between them and ratios of such counts. Comparison with observation nevertheless requires quantities reported in SI units. This part states how the two are kept apart, lists the symbols of the theory, and states which numbers are parameters and which are bridges.

25 Three descriptive layers

The theory separates three layers of description. The separation is a structural requirement, not a convenience.

layer	role	permitted content
causal layer	physical ontology and dynamics	events, the relation $\prec$, counts of events, flows, response variables, the action of the order
bridge layer	metrology only	the discreteness scale, unit conversion, projection of counts onto lengths, times and accelerations
observational layer	empirical comparison	SI quantities, catalogues, measured accelerations, delays, deflections

Three rules follow. Causal variables never carry physical dimensions. Observational quantities never take part in causal evolution. Bridge parameters never change a rule, a threshold or a weight of the causal layer. A construction that violates any of the three reintroduces background structure and is excluded.

Coordinates have a special status. They are not part of the causal layer. An *instrument* may use them as a declared readout, for example to place the events of a sprinkling or to report a profile as a function of radius; a *rule* of the dynamics may not (Sec. 29).

26 Catalogue of symbols

For Greek letters and special symbols the middle column gives the name and the pronunciation (International Phonetic Alphabet, English).

26.1 Structure of the order

symbol	read as	meaning
$v,\ u,\ x,\ y$		events (elements of the causal set)
$u \prec v$	u precedes v /prɪˈsiːdz/	causal relation: u can influence v
$u \prec\!\cdot\, v$	link /lɪŋk/	link: $u \prec v$ with nothing in between
$I(u,v)$		causal interval $\{w : u \prec w \prec v\}$
$n(u,v)$		number of elements strictly inside $I(u,v)$
$J^-(x)$	J minus /dʒeɪ ˈmaɪnəs/	causal past of x
ρ	rho /roʊ/	**fundamental density of the substrate**: the mean number of events per unit proper four-volume. In a sprinkling the number of events in a region of volume V is Poisson distributed with mean ρV. It sets the discreteness length $\ell = \rho^{-1/D}$ and, with ϵ, the non-locality scale ℓ_k. It is the same everywhere: near matter the *coordinate* density changes (ρ_c), not ρ
$D = d_s + 1$		spacetime dimension; d_s spatial dimensions
N		number of events in a region

26.2 Transport, matter and action

symbol	read as	meaning
F_v		flow carried by event v (non-negative, conserved)
ΔQ_v	delta /ˈdɛltə/ Q	flow stored at v (the residual of the balance at v)
s_v		source term at v
u^a		four-velocity carried by a flow as its state
$P^a = F\,u^a$		four-momentum of a flow; merging rule $F = \lvert P\rvert$
T^{ab}		energy–momentum tensor of flows, Eq. (20.1)
e_0		rest-frame energy density of a flow ($T_{ab}u^a u^b$)
C		capacity: the largest flow an event can pass on
$\sigma_v = F_v/C$	sigma /ˈsɪgmə/	load of event v
$\Delta\tau$	delta tau /ˈdɛltə taʊ/	proper time between successive events of a flow, counted by the order
$S_m = -\sum m\,\Delta\tau$	Σ: sum /sʌm/	action of a flow (Axiom 8)
$\mathcal{R} = \sum_v Z_v\,\Delta\tau$	script R /skrɪpt ɑːr/	Causal Resistance Functional, the impedance-weighted action

26.3 Response of the substrate

symbol	read as	meaning
Π_v, $0 < \Pi_v \le 1$	capital pi /ˈkæpɪtəl paɪ/	permeability, read as the local clock rate: $\mathrm{d}\tau = \Pi\,\mathrm{d}t$ along a static world line
$L = \ln\Pi$	log pi /lɑg paɪ/	logarithmic clock potential; the stress coordinate of the substrate ($L < 0$ near matter)
$I_v \ge 1$		stiffness: resistance of the order to reconfiguration
$Z_v = I_v/\Pi_v \ge 1$		impedance
c_{eff}	c effective /siː ɪˈfɛktɪv/	local speed of the causal cone; $c_{\text{eff}} = c$ in the empty substrate
ρ_c	rho c /roʊ siː/	density of events per *coordinate* volume, in units of ρ ($= \sqrt{-g}$ in a sprinkling); grows near matter
ρ_s	rho s /roʊ ɛs/	density of saturated events: the source of the static field in the rest frame of the matter ($T_{ab}u^a u^b$ of the flows)
$q = \rho_c\,c_{\text{eff}}$		causal throughput (conserved, Sec. 22)

26.4 Geometry read from the order

symbol	read as	meaning
ϕ	phi /faɪ/	clock potential, read from the number of elements of a small interval at rest
ψ	psi /saɪ/	spatial potential; $\phi+\psi$ is read from the boundary of the relation and $\phi-d_s\psi$ from the density
$\gamma=\psi/\phi$	gamma /ˈgæmə/	post-Newtonian parameter (light deflection, Shapiro delay)
β	beta /ˈbeɪtə/	post-Newtonian parameter of the non-linearity of the clock field
U		Newtonian potential ($U=-\phi$ in the weak field)
η_{ab}	eta /ˈeɪtə/	Minkowski metric
h_{ab}, $\bar{h}_{ab}$	h bar /eɪtʃ bɑːr/	metric perturbation and its trace-reversed form
$\Box_g$	box /bɑks/, d'Alembertian	wave operator of the metric g
R		scalar curvature

26.5 Operators and the action of the order

symbol	read as	meaning
ϵ, $0<\epsilon\le 1$	epsilon /ˈɛpsəlɑn/	smearing parameter of the Benincasa–Dowker operator
$\ell_k=(\epsilon\rho)^{-1/D}$	ell k /ɛl keɪ/	non-locality scale
C_i		Benincasa–Dowker coefficients; in 3+1 $(-1,9,-16,8)$, fixed by D
a_D		normalisation; $a_4=4/\sqrt{6}$
$K_\epsilon(n)$	K epsilon /keɪ ˈɛpsəlɑn/	smeared kernel, Eq. (24.1)
$g_k(w)$		expected kernel, Eq. (24.4)
B_ϵ, $\mathbb{E}[B_\epsilon]$	expectation of B /ɛk.spɛkˈteɪʃən/	discrete BD operator and its expectation
S_{BD}		BD action of the order, Eq. (24.5)
$V(x,y)$		proper volume of the interval between y and x
$\hat{K}(k,T)$	K hat /keɪ hæt/	ratio of the second-order BD action to its Einstein–Hilbert value (Sec. 35); K_{loc} its local part
c_T		coefficient of the age term of the BD action
$\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$	script A, B, C /skrɪpt eɪ biː siː/	coefficients of the static quadratic form of the action of the order: conformal, mixed and optical sector (Einstein–Hilbert 6, 8, 2), Sec. 36.3
T_{U}		Unruh temperature (Sec. 38); not the age T
T		age: how long a static field has existed in the past of an event
κ	kappa /ˈkæpə/	coupling of flows to the order
$\hbar$	h-bar /eɪtʃ bɑːr/	reduced Planck constant
ℓ_P	ell P /ɛl piː/	Planck length

27 Parameters of the theory

Fundamental. The causal layer has two numbers that are not fixed by the order itself: the density ρ, which sets the discreteness length $\ell=\rho^{-1/D}$, and the smearing parameter ϵ, which sets the non-locality scale $\ell_k=(\epsilon\rho)^{-1/D}\ge\ell$. All results of Part V are stated in units of ℓ_k;

the expectation values of the operator and of the action depend on ϵ and ρ only through the product $\epsilon\rho$. For $\epsilon = 1$, $\ell_k = \ell$.

Fixed by the dimension. The coefficients C_i and the normalisation a_D are not parameters: they are fixed by requiring that the expected operator reproduce the d'Alembertian in D dimensions.

The coupling. The strength with which a flow acts on the order, κ, becomes Newton's constant in the bridge layer. It is not tuned in the causal layer: in the linear regime it fixes only the normalisation of the field, never its shape or its tensor structure.

Working parameters of models. Particular models of the response of the substrate (the saturation level of a load-driven clock law, a materialisation threshold, a feedback exponent) carry working parameters. They are properties of a model, not constants of nature, and no result of this paper depends on their values.

Laboratory units. The laboratory works in units in which $\rho = 1$, or $\epsilon\rho = 1$ for computations with the expected operator; lengths are then counted in ℓ or ℓ_k.

28 The SI bridge

The speed of light. In a Lorentz-invariant substrate the boundary of the relation has the same slope in every frame. Reporting that slope in SI units defines c; c is a conversion between the counting of events along time and across space, not an additional constant of the causal layer.

Newton's constant (conditional). The bridge to G goes through the entropy of horizons (Sec. 38). If the entropy of a horizon is the number of horizon molecules times an entropy s_0 per molecule, the entropy density per unit area is $\eta = s_0\, a^{(4)}/\ell^2$ with $a^{(4)} = \sqrt{3}/10$, and the thermodynamic derivation of Einstein's equations gives $G = 1/(4\hbar\eta)$ [7]. Then

$$\ell^2 = 4 s_0\, a^{(4)}\, \ell_P^2 = \frac{2\sqrt{3}}{5}\, s_0\, \ell_P^2, \qquad \ell \simeq 1.075\sqrt{s_0}\, \ell_P, \tag{28.1}$$

with ℓ_P the Planck length. The discreteness length is thereby tied to G up to s_0. This is a relation of consistency, not a prediction of G: it becomes predictive only when ℓ or s_0 is measured independently.

The non-locality scale. The smearing scale ℓ_k is not fixed by any bridge. It is a free scale of the theory, to be bounded or measured by the predictions of Part VII. For fields propagated by the non-local operator existing data give $\ell_k \lesssim 10^{-19}$ m; whether this applies to Causal Physics is discussed in Sec. 45.

Projection of counts. Within the plateau, where geometry is a reliable compression of the order, counts are projected onto SI quantities: a number of elements N of a region onto a four-volume N/ρ, the length of the longest chain onto a proper time, the clock potential L onto the Newtonian potential U/c^2. Outside the plateau the projections persist formally but lose physical meaning.

No acceleration scale. The present formulation introduces no acceleration scale in any layer (Sec. 42).

Part IV
Axioms of Causal Physics

This part states the foundations of Causal Physics in two groups. The *methodological principles* say how the theory may be built and tested; they are rules of inquiry and say nothing about nature. The *physical axioms* say what the world is made of and how it behaves; they are the content of the theory. For each physical axiom we state its status: whether it is an *input* (assumed), a *derived* statement, or *tested* by the laboratory.

No background manifold, metric, coordinate system or spacetime dimension is assumed. All such structures, where they exist, are read from the order.

29 Methodological principles

Definition 29.1 (M1: Causal primacy)**.** No geometry, coordinate system, external clock or unit enters the causal layer. Geometry is a description of the order, read from it by declared instruments.

Definition 29.2 (M2: Separation of layers)**.** Causal, bridge and observational quantities are kept apart (Sec. 25). Observational quantities never act on the causal dynamics, and bridge parameters never change its rules.

Definition 29.3 (M3: Order and number)**.** Every rule of the dynamics uses only the order and counts of events, or states of events derived from them. Coordinates may appear in an instrument as a declared readout, never in a rule.

Definition 29.4 (M4: Causality and covariance of rules)**.** The state of an event depends only on its causal past. A rule gives the same result in every inertial frame and in every order of computation compatible with the causal order (every linear extension).

Definition 29.5 (M5: Finiteness)**.** A rule and its results are finite and well defined without a wall of the region or a regulator. A result that holds only because of the boundary of a simulation is a property of the simulation.

Definition 29.6 (M6: Input and output)**.** Every statement is classified as an input (postulate, tuned parameter) or an output (it followed without being put in). A test of an output that was put in as an input proves nothing.

Definition 29.7 (M7: Pre-registration and falsifiability)**.** Every test states its question, prediction, null model and numerical criterion before it is run, and declares the level of conclusion it can support: a *calibration* establishes an instrument and says nothing about physics; a *mechanism* shows that a mechanism works in a declared model; a *chain* shows that connected mechanisms reproduce a predicted behaviour; a *prediction* is an independent result fixed in advance. No output may claim more than its level allows. A changed criterion makes a new revision of the test, and the earlier verdict stays. A new hypothesis may follow a failure, but it may not change the criteria by which the earlier test failed. Negative results are kept.

30 Physical axioms

30.1 The substrate

Axiom 1 (Primacy of events)**.** Physical reality consists of elementary events. An event is an irreducible act of realisation, not a persisting entity. It has no intrinsic position, duration, mass, energy, charge or scale. All physical attributes are relational: they are patterns of causal connectivity among events.

Status. Input.

Axiom 2 (Causal order)**.** Causal relations form a partial order: they are transitive and acyclic, and every interval contains finitely many events. Closed causal loops do not exist. Time is the order itself; there is no external time and no global simultaneity.

Status. Input. The structure is a causal set.

Axiom 3 (Lorentz-invariant substrate)**.** On the plateau the order is, statistically, a uniform sprinkling of a Lorentzian manifold at density ρ: the number of events in a region is ρ times its proper volume, and no frame is preferred. Dimension, distance and duration are read from order and number alone.

Status. Input, supported by tests: the degree of an event is not bounded (F0.3), local growth rules with bounded degree produce no manifold-like order (F6.1, F6.2), and geometry is read blindly from the order (K.1, K.3, K.16b, K.16c).

30.2 Transport and matter

Axiom 4 (Causal continuity)**.** Influence is transported along the relations of the order and is conserved at every event:

$$\sum_{\text{in}} F + s_v = \sum_{\text{out}} F + \Delta Q_v. \tag{30.1}$$

What enters an event leaves it or is stored there.

Status. Input. Tested consequence: Gauss's law for a steady source, $J/S = 1.003$–1.004 in 2+1 and 3+1 (F1.1).

Axiom 5 (Finite capacity)**.** The flow an event can pass on is bounded by a capacity C. Capacity limits the flow through an event, not the number of its relations.

Status. Input. Supported by F1.8 (not yet robust to the seed).

Axiom 6 (Flows carry momentum)**.** A flow carries its four-velocity u^a as a state passed from event to event. When flows merge, the merged flow is the length of the summed four-momentum, $F = |\sum P|$, so that four-momentum is conserved. The flows then have an energy–momentum tensor T^{ab}, Eq. (20.1).

Status. Input. Tested consequences: a flow with a carried direction is localised and moves with its source, i.e. has inertia (F3.6, replicated); T^{ab} is measured correctly for dust, counter-streaming beams and an isotropic gas, and the merging rule conserves four-momentum exactly (K.15, replicated).

30.3 Response and action

Axiom 7 (Local response)**.** Every event has a response variable Π_v, $0 < \Pi_v \leq 1$, read as its clock rate, and a stiffness $I_v \geq 1$; its impedance is $Z_v = I_v/\Pi_v$. The axiom does not fix how Π_v depends on the load, on the surrounding order or on other causal variables; that dependence is a separate, independently testable law of response.

Status. Input. The law of response is not part of the axiom. In the linear regime it is fixed by the dynamics of the order (Axiom 9, Part V).

Axiom 8 (Action of a flow)**.** The action of a flow is its proper time counted by the order,

$$S_m = -\sum m\,\Delta\tau, \tag{30.2}$$

where $\Delta\tau$ is the proper time between successive events of the flow, as given by the length of the longest chain or by the number of elements of a small interval. The Causal Resistance Functional $\mathcal{R} = \sum_v Z_v \Delta\tau$ is its impedance-weighted form.

Status. Input. Derived consequences: on a fixed order the motion is geodesic in the conformal geometry $\Pi^2\eta$ (tested: F2.2, F2.4, F2.5); the coupling of a flow to the order is $\delta S_m/\delta h_{ab} = \frac{1}{2}T^{ab}$, i.e. through the rest frame of the flow, and contributions of different flows add (Sec. 21). How the weight Z_v relates to the response of the order beyond the linear regime is open.

30.4 Dynamics of the order

Axiom 9 (The order responds to matter)**.** The order is dynamical. Its configurations are weighted by an action of the order, and matter changes the order: near flows, causal cones narrow and events are added. Causality itself is preserved (Axioms 2 and 4). The candidate action of the order is the Benincasa–Dowker action S_{BD} (Sec. 24).

Status. Input (that the order is dynamical and which action weights it). Derived and computed consequences:

- the expected BD action has the Einstein–Hilbert stiffness in the static conformal sector, 6.0003 against 6 (F6.8a), in the mixed sector, 7.983 against 8 (F6.8b2), and in the optical sector, 2.002 against 2 (F6.8c2); with Axiom 8 the linear response of the order is then the coupling of linearised general relativity to T_{ab} (Part V);
- in 3+1 this response is the conservation of causal throughput, $\rho_c\, c_{\mathrm{eff}} = \text{const}$, which gives $\gamma = 1$ (tested on discrete orders: F7.1c, raw $\gamma = 1.000 \pm 0.045$, formal verdict undecided);
- a substrate rule realising it makes pressure and motion gravitate and drags the cones (tested: F7.4a).

How the order grows (a sequential growth law or a sum over orders weighted by S_{BD}) is open.

31 What is derived, and from what

statement	from	status
time as order, no global simultaneity	Ax. 2	derived
geometry read from order and number	Ax. 3	tested (K.16b/c)
Gauss's law	Ax. 4	tested (F1.1)
inertia of localised matter	Ax. 6	tested (F3.6)
conservation of four-momentum in merging	Ax. 6	tested (K.15)
geodesic motion	Ax. 8	tested (F2)
coupling through T_{ab} in the rest frame of each flow	Ax. 8	derived
Einstein–Hilbert stiffness of the order	Ax. 9	computed, all three sectors (F6.8a, F6.8b2, F6.8c2)
throughput principle, $\gamma = 1$ in 3+1	Ax. 8, 9	derived; F7.1c
tensor coupling, frame dragging	Ax. 8, 9	derived; F7.4a
non-linear completion by general relativity	Ax. 9 + uniqueness	derived (programme)

Part V
The Geometric Limit: Gravity from the Response of the Order

This part derives and tests the gravitational coupling of Causal Physics in the regime where geometry is a reliable description of the order. It proceeds in five steps: what the coupling must look like (Sec. 32), a substrate rule that realises it on causal sets (Sec. 33), why the rule follows from the action of the order (Sec. 34), a direct computation of that action to second order (Sec. 35), and the static propagator that turns the result into predictions (Sec. 36). The static section of the field to first post-Newtonian order (Sec. 37) and the thermodynamic completion beyond the linear order (Sec. 38) close the part.

Conventions: signature $(-,+,+,+)$, $c = 1$, $g = \eta + h$, $\bar{h}_{ab} = h_{ab} - \frac{1}{2}\eta_{ab}h$. A static weak field has $h_{00} = 2U$, $h_{ij} = 2\gamma U\delta_{ij}$, with $U > 0$ the Newtonian potential and u^a the four-velocity of the source.

32 What the coupling must look like

32.1 The parameter γ is a ratio of two couplings

In the rest frame of a dust source, a static weak field with parameter γ in d_s spatial dimensions has

$$\bar{h}_{00} = U(1 + \gamma d_s), \qquad \bar{h}_{ij} = U(1 + 2\gamma - \gamma d_s)\,\delta_{ij}. \tag{32.1}$$

In 3+1 this is

$$\bar{h}_{ab} = U\big[(1-\gamma)\,\eta_{ab} + 2(1+\gamma)\,u_a u_b\big]. \tag{32.2}$$

The field is a mixture of two couplings: to the trace of the energy–momentum tensor (η_{ab}) and to the tensor itself ($u_a u_b$).

- $\gamma = -1$: $\bar{h}_{ab} = 2U\eta_{ab}$, pure trace coupling (spin 0, Nordström). Light is not deflected and does not gravitate, because the electromagnetic field has zero trace. This is the geometry

of a fixed order (Sec. 21).

- $\gamma = 1$: $\bar{h}_{ab} = 4U\, u_a u_b$, pure coupling to $T_{ab} = \rho\, u_a u_b$ with no trace part. This is linearised general relativity, $\Box \bar{h}_{ab} = -16\pi G\, T_{ab}$.
- $\gamma = 1/3$ (cones that only narrow) is a mixture of the two.

32.2 The throughput principle is the tensor coupling

The throughput principle $\rho_c\, c_{\text{eff}} = \text{const}$ (Sec. 22) gives $\gamma = 2/(d_s - 1)$. Pure tensor coupling of dust ($\bar{h}_{ij} = 0$) gives $\gamma = 1/(d_s - 2)$. The two coincide only for $d_s = 3$. In three spatial dimensions, therefore, the throughput principle *is* the coupling of linearised general relativity: the deformation of the order lies only along the flow ($u_a u_b$) and vanishes across it. Three consequences follow at once:

- light gravitates, and pressure gravitates as $\rho + 3p$;
- a moving source produces g_{0i}: inertial frames are dragged;
- the field of many flows is a sum over flows, $\bar{h}_{ab} \propto \sum F\, u_a u_b\, G_{\text{ret}}$, whose transverse traceless part is a gravitational wave with the two polarisations $+$ and $\times$.

32.3 A scalar field cannot be the whole law

A single scalar clock field Π in a fixed frame satisfies the static weak-field tests if the metric is chosen with $\gamma = 1$, but it cannot be the complete law of gravity. It has no g_{0i}, so it cannot reproduce the frame dragging measured by Gravity Probe B [18]. Normalised to Newton's law, a scalar field radiates quadrupole power $P = \frac{G}{60}\big[(\text{tr}\,\dddot{I})^2 + 2\,\dddot{I}_{ij}\,\dddot{I}_{ij}\big]$, which for a circular binary is exactly 1/6 of the general relativistic value; binary pulsars agree with general relativity to 10^{-3}–10^{-4} [19]. A scalar field is therefore the static section of the field in the rest frame of its source, not the field itself (Sec. 37).

33 The substrate rule and its test

33.1 The rule

The coupling of Sec. 32 requires the order to know in which frame each flow is at rest. The flows carry this information (Axiom 6). The substrate rule is:

1. *own frame*: every flow narrows the causal cone in its own rest frame;
2. *additivity*: flows that load the same relation share its capacity, and their cone deficits add;
3. *throughput*: the density of events rises by $e^{\sigma_{\text{tot}}}$, with σ_i the rest-frame load of flow i and $\sigma_{\text{tot}} = \sum_i \sigma_i$.

For a candidate relation with separation d^a from an event loaded by flows with four-velocities u_i, the relation is kept if and only if

$$d^0 > 0, \qquad -d_0^2 + |\boldsymbol{d}|^2 + \sum_i a_i\,(u_i \cdot d)^2 \le 0, \qquad a_i = \frac{\sigma_i}{\sigma_{\text{tot}}}\big(1 - e^{-2\sigma_{\text{tot}}}\big). \tag{33.1}$$

For flows at rest this is the single-flow throughput rule with cones narrowed to $e^{-\sigma_{\text{tot}}}$. The order is built from a flat sprinkling by keeping or discarding relations event by event; no metric is sprinkled, and no field equation, Green function or trace reversal enters.

33.2 The test: counter-streaming flows

Two equally strong flows moving at $+v$ and $-v$ have zero momentum density but non-zero energy and spatial stress. The candidate rules therefore predict different clock fields. With the rest-frame load of each flow fixed, the amplitude of the clock field relative to $v = 0$ is predicted to be:

rule	linear prediction	at $v = 0.6$
tensor rule (33.1)	$\Gamma_v^2(1 + v^2)$	2.125 (exact, non-linear: 2.361)
energy density in the mean-flow frame	Γ_v^2	1.5625
laboratory-frame density	Γ_v	1.25
rest density (event count)	1	1

($\Gamma_v = 1/\sqrt{1 - v^2}$.) The amplitude is read in the near field, transverse to uniform flows, because for a compact stationary system the stress integrates to zero (von Laue) and the $1/r$ tail cannot distinguish the rules.

Result (F7.4a, 3+1, 16 sprinklings per configuration, $\sim 5.6 \times 10^4$ events, PASSED). The blind instrument reads:

- amplitude ratio $R = A(0.6)/A(0) = 2.401 \pm 0.053$ against 2.361 for the tensor rule; the energy density reading (1.5625) is 16 standard errors away. Pressure gravitates, not only energy;
- cone anisotropy along/across the flows $K = 1.221 \pm 0.012$ against 1.192; every scalar rule of the mean-flow frame gives exactly 1;
- a single flow at $0.6c$ (reported): $R = 2.538$ against 2.519; the cone tilts, with a deficit against the flow 17.6 times the deficit along it (prediction 12, scalar rules 1). The order drags its cones with the flow. The downstream deficit matches exactly (-0.065 against -0.065).

The flat control reads zero, and an isotropic order reads isotropic ($K = 1.002 \pm 0.004$) (Fig. 4).

The test shows that a substrate with this rule couples to T_{ab} as general relativity does at the linear order, including frame dragging, with the tensor structure arising from the response of the order alone. The rule itself is an input. The next section shows that it is not an independent one.

34 The rule from the action of the order

34.1 The matter side

By Axiom 8 the action of a flow is its proper time counted by the order. For $g = \eta + h$,

$$\frac{\delta S_m}{\delta h_{ab}} = \tfrac{1}{2}\, T^{ab}. \tag{34.1}$$

Two parts of the substrate rule follow without any knowledge of the dynamics of the order:

- *own frame*: the order sees a flow only through $u^a u^b$, i.e. in the frame in which its proper time is measured;
- *additivity*: the action is a sum over flows, so at the linear order their contributions add.

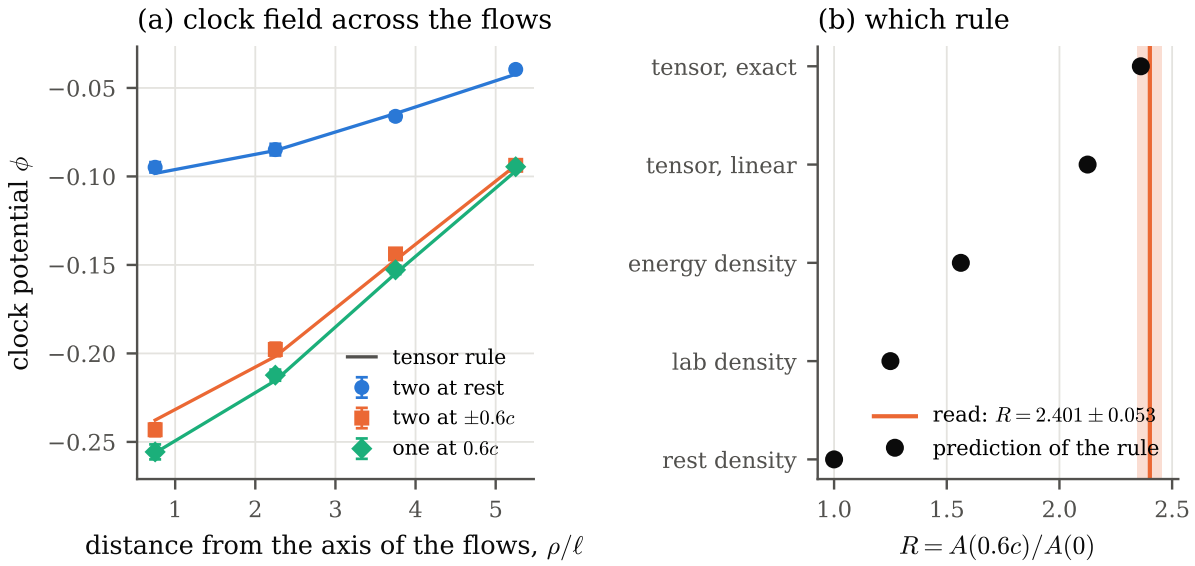


Figure 4: Counter-streaming flows (F7.4a, 3+1, 16 sprinklings per configuration). (a) The clock potential read across the flows (points, ±1 standard error) against the tensor rule (lines) for two flows at rest, two flows at $\pm 0.6c$ and one flow at $0.6c$. (b) The amplitude ratio $R = A(0.6c)/A(0)$ read (orange line, band ±1 standard error) against the prediction of each candidate rule.

34.2 One number is left

A linear, local and Lorentz-invariant response $h_{ab} = G * T$ has two channels, T_{ab} and $\eta_{ab}T$. For static dust this is the one-parameter family (32.2): the structure of the rule is forced up to γ. The number γ is fixed by the action of the order. For a general static quadratic action coupled to dust,

$$\mathcal{L} = c_{\Phi\Phi}|\nabla\Phi|^2 + c_{\Phi\Psi}\nabla\Phi\cdot\nabla\Psi + c_{\Psi\Psi}|\nabla\Psi|^2 - \kappa'\rho\,\Phi, \tag{34.2}$$

with $\mathrm{d}s^2 = -(1+2\Phi)\mathrm{d}t^2 + (1-2\Psi)\mathrm{d}\boldsymbol{x}^2$, the variation with respect to Ψ gives

$$\gamma = \frac{\Psi}{\Phi} = -\frac{c_{\Phi\Psi}}{2\,c_{\Psi\Psi}}, \tag{34.3}$$

independently of $c_{\Phi\Phi}$. The Einstein–Hilbert action gives (computed symbolically for $d = 2, 3, 4$)

$$c_{\Phi\Phi} = 0, \qquad c_{\Phi\Psi} = -2(d-1), \qquad c_{\Psi\Psi} = (d-1)(d-2) \quad \Rightarrow \quad \gamma = \frac{1}{d-2}, \tag{34.4}$$

which is $\gamma = 1$ in 3+1. Two remarks:

- $c_{\Phi\Phi} = 0$: clocks have no stiffness of their own; the time part of the field is fixed by the spatial part and the source.
- The throughput principle is a consequence: dust has no stress and the order has the Einstein–Hilbert stiffness, so the trace-reversed spatial deformation vanishes.

34.3 The chain

With the Benincasa–Dowker action as the dynamics of the order (Axiom 9):

$$\begin{gathered}\text{proper time counted by the order} \;+\; \text{BD action of the order} \\ \Longrightarrow \;\text{the substrate rule at the linear order } (\gamma = 1),\end{gathered}$$

provided the BD action has the Einstein–Hilbert stiffness. It is known that the expectation of the BD action tends to the Einstein–Hilbert action, with a boundary term, in the continuum

limit [4, 20]. The quantity that matters here, however, is the static quadratic form of the *discrete* action at a finite non-locality scale, including the contributions of long, nearly null intervals. The next section computes it.

35 The Benincasa–Dowker action to second order

35.1 Sectors

Write a static metric as $g = \Omega^2[-\mathrm{d}t^2 + n^2 \mathrm{d}\boldsymbol{x}^2]$ with $\omega = \ln\Omega$ and $\nu = \ln n$. The conformal factor ω does not change the causal order, only the density of events; the optical index ν changes the light cones. For plane waves $\omega = a\cos kz$, $\nu = b\cos kz$ the Einstein–Hilbert action, averaged over a period, is (symbolic computation)

$$\left\langle \sqrt{-g}\, R \right\rangle = \frac{k^2}{2}\left(6a^2 + 8ab + 2b^2\right) + O(a^3, b^3). \tag{35.1}$$

Matter couples only to $\omega = \Phi$; in these variables $\gamma = -1 + \mathcal{B}/(2\mathcal{C})$ with $\mathcal{B} = 8$, $\mathcal{C} = 2$. We compute here the conformal coefficient, 6 in Einstein–Hilbert, which is an exact quadrature because the order is unchanged.

35.2 Method

For $g = e^{2af}\eta$ with static f, the expected action per unit time and volume in the bulk of a region of age T (the past of every event cut at time T) is

$$\mathbb{E}[S] = -2\big\langle \sqrt{-g}\; \mathbb{E}[B_\epsilon 1](x)\big\rangle, \tag{35.2}$$

with $\mathbb{E}[B_\epsilon 1]$ from Eq. (24.3). The proper volume of an interval is the flat volume times the average of e^{4af} over the interval. The second derivative with respect to a at $a = 0$ is taken analytically under the integral: with $P = e^{4af}$, $\langle\cdot\rangle$ the average over the interval, $g_1 = w\, g_k'(w)$ and $g_2 = w^2 g_k''(w)$,

$$\frac{\partial^2}{\partial a^2}\Big[P_x P_y\, g_k\big(w\langle P\rangle\big)\Big]_{a=0} = 16\Big[(f_x + f_y)^2 g_k + 2(f_x + f_y)\langle f\rangle g_1 + \langle f\rangle^2 g_2 + \langle f^2\rangle g_1\Big]. \tag{35.3}$$

The field is a static plane wave $f = \cos kz$, averaged over its phase in closed form, so that no wall and no slowly converging tails enter. The averages of a plane wave over a causal interval are the Fourier transform of the 3+1 causal diamond, which we use in closed form: in the rest frame of the interval, with half-height h and four-momentum (ω', q'),

$$F(\omega', q') = \frac{2}{V}\int_0^h \cos\big(\omega'(h-u)\big)\,\frac{4\pi\,(\sin q'u - q'u\cos q'u)}{q'^3}\,\mathrm{d}u, \qquad V = \frac{2\pi h^4}{3}, \tag{35.4}$$

evaluated stably for nearly null momenta. Lengths are in units of ℓ_k ($\epsilon\rho = 1$); the expectation depends on ϵ and ρ only through their product. The result is reported as

$$\hat{K}(k, T) = \frac{\partial_a^2 \mathbb{E}[S]}{\text{Einstein–Hilbert value } 6k^2}. \tag{35.5}$$

The Einstein–Hilbert number enters only this normalisation.

Instruments (all within tolerance): the closed-form transform agrees with an independent quadrature of the diamond to 7×10^{-8}; the analytic second derivative agrees with finite differences of the curved operator to 3×10^{-4}; a constant conformal factor gives zero response to 5×10^{-5} of the signal; doubling every quadrature parameter changes $\hat{K}$ by at most 3×10^{-5}.

35.3 Results

For every wavenumber the response is linear in the age T (residuals $\leq 10^{-3}$), so that

$$\hat{K}(k,T) = K_{\rm loc}(k) - c_T\, k^3\, T. \tag{35.6}$$

$k\ell_k$	0.025	0.035	0.05	0.07	0.1	0.14	0.2	0.3	0.5
$K_{\rm loc}$	0.9904	0.9826	0.9680	0.9438	0.8994	0.8287	0.7056	0.4687	−0.015

The local part. A fit $K_{\rm loc} = K_0 - k^2[c_1 \ln(1/k) + c_0]$ over $k\ell_k \leq 0.1$ gives

$$K_0 = 1.00005, \qquad c_1 = 3.98, \qquad c_0 = 0.91, \qquad \text{rms } 2.3\times 10^{-5}. \tag{35.7}$$

The conformal stiffness of the discrete BD action is therefore $6K_0 = 6.0003$: *the Einstein–Hilbert value emerges from the counting of intervals* (F6.8a, PASSED). The discreteness corrects it non-analytically,

$$1 - K_{\rm loc} \simeq (k\ell_k)^2\big[4.0\ln(1/k\ell_k) + 0.9\big], \tag{35.8}$$

and the stiffness changes sign at $k\ell_k \simeq 0.49$, a wavelength of $12.7\,\ell_k$. The momentum-space form factor of the continuum non-local d'Alembertian is known in closed form [21, 22]; a quantitative comparison of its infrared expansion with the non-analytic correction found here is a planned independent check.

The age term. The second term of Eq. (35.6) is not local: the static response weakens in proportion to how long the field has existed. We predicted it before computing it. A long, nearly null interval running perpendicular to $\boldsymbol{k}$ stays in phase with the field over its whole length. Expanding its contribution in the proper time τ of the interval, the moment conditions of the BD kernel, $\int u^p g_k(u)\,\mathrm{d}u = 0$ for $p = -\frac{1}{2}, 0, \frac{1}{2}$, remove the orders τ^0, τ^2 and τ^4, but not τ^6. Each unit length of history therefore adds a contribution of order k^5, which is $k^3 T$ relative to k^2. In the limit of long intervals the contribution depends only on the phase $\phi = kL\cos\theta$ and on $k\tau$, and the coefficient is a two-dimensional integral of the diamond transform,

$$\frac{\mathrm{d}J}{\mathrm{d}T} = \frac{2\pi}{k}\int \mathrm{d}\phi \int \tau\,\mathrm{d}\tau \left[2\cos\tfrac{\phi}{2}\, F\, g_1 + \tfrac{1}{2}F^2 g_2\right], \tag{35.9}$$

which gives $c_T = 0.1012$–0.1022 for $k\ell_k = 0.05$–0.2. The full computation gives $c_T = 0.1035 \pm 0.0003$ for $k\ell_k \leq 0.2$ and 0.098–0.104 over the whole range: agreement to 2%, the rest being the finite age. In physical units the relative size of the age term is

$$0.1\,(k\ell_k)^3\,\frac{T}{\ell_k} = 0.1\,k^3\ell_k^2\,T. \tag{35.10}$$

Figure 5 shows the local part with its fit and the collapse of the age term.

36 Covariance and the static propagator

36.1 Form factors

The expected BD action depends only on the geometry; it is exactly invariant under diffeomorphisms. If its quadratic part around flat space is also covariant, it has the form

$$S_2 = \int \sqrt{-g}\left[R + R\,F_0(\Box)\,R + C_{abcd}\,F_2(\Box)\,C^{abcd}\right], \tag{36.1}$$

the basis $\{R^2, C^2\}$ being complete at this order. For the plane waves of Eq. (35.1), $R^{(1)} = 2k^2(3a+2b)\cos kz$ and $C^{(1)}\cdot C^{(1)} = \frac{4}{3}\,b^2k^4\cos^2 kz$: the Weyl tensor comes from the optical index alone. Per $k^2/2$ the three coefficients become

$$\mathcal{A} = 6 + 9\varphi_0, \qquad \mathcal{B} = 8 + 12\varphi_0, \qquad \mathcal{C} = 2 + 4\varphi_0 + \varphi_2, \qquad \varphi_0 = 4k^2F_0, \quad \varphi_2 = \tfrac{4}{3}k^2F_2. \tag{36.2}$$

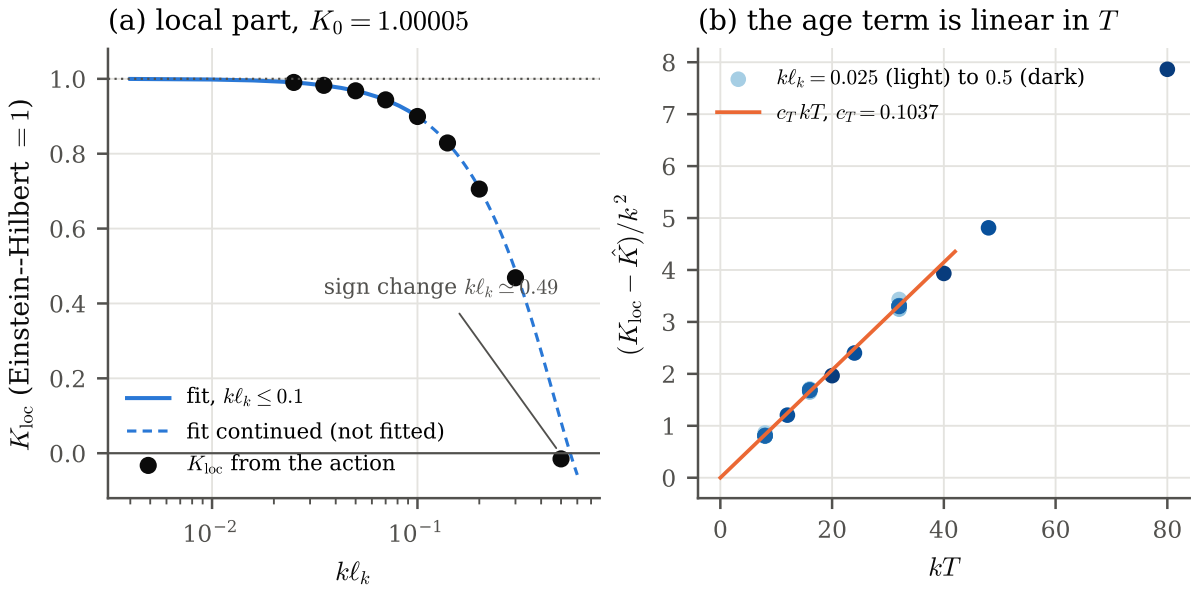


Figure 5: The conformal sector (F6.8a). (a) The local part $K_{\rm loc}$ of the stiffness in units of the Einstein–Hilbert value (points), the fit $K_0 - k^2[c_1 \ln(1/k) + c_0]$ over $k\ell_k \le 0.1$ (solid) and its continuation (dashed), which the points beyond the fitted range follow to the sign change at $k\ell_k \simeq 0.49$. (b) The age term: $(K_{\rm loc} - \hat{K})/k^2$ for every computed age and wavenumber (light to dark: $k\ell_k = 0.025$ to 0.5) against kT, with the line $c_T\,kT$, c_T the mean of the per-k slopes over $0.05 \le k\ell_k \le 0.2$. The two largest wavenumbers leave the line at large kT.

36.2 The field of a point mass

Matter couples to $\omega = \Phi$. Solving the quadratic form with a source gives the standard decomposition into spin 2 and spin 0:

$$\frac{\Phi}{\Phi_{\rm GR}} = \tfrac{4}{3}S_2(k) - \tfrac{1}{3}S_0(k), \qquad \frac{\Psi}{\Psi_{\rm GR}} = \tfrac{2}{3}S_2(k) + \tfrac{1}{3}S_0(k), \qquad \gamma(k) = \frac{2S_2 + S_0}{4S_2 - S_0}, \tag{36.3}$$

with $S_0 = 1/(1 + \frac{3}{2}\varphi_0)$ and $S_2 = 1/(1 - \frac{3}{2}\varphi_2)$. Since $1 + \frac{3}{2}\varphi_0 = \mathcal{A}/6 = \hat{K}$, *the computation of Sec. 35 determines the whole spin-0 channel*:

$$S_0(k) = \frac{1}{\hat{K}(k,T)} = \frac{1}{K_{\rm loc}(k) - c_T\,k^3 T}. \tag{36.4}$$

The spin-0 propagator has a static pole where $\hat{K} = 0$: at $k\ell_k \simeq 0.49$ without the age term, and with it at the length scale $(0.1\,\ell_k^2 T)^{1/3}$.

These expressions assume that the quadratic action is covariant at every k. Section 36.3 tests this assumption directly.

36.3 The mixed sector

Covariance makes a sharp prediction without a free parameter: the mixed coefficient follows the conformal one, $\mathcal{B}/8 = \mathcal{A}/6 = \hat{K}$ at every k. We computed $\mathcal{B}$ directly (F6.8b2), for $g = e^{2af}\big(-\mathrm{d}t^2 + e^{2bh}\mathrm{d}\boldsymbol{x}^2\big)$ with $f = h = \cos kz$, phase-averaged.

Method. The optical index changes the order. To first order in b, y precedes x if and only if their time separation exceeds the optical length of the straight segment between them; by Fermat's principle the change of the path is of second order. The past of x is parametrised on

the unperturbed cone. The volume of an interval is the integral, over the spatial projection of the diamond (a prolate spheroid, integrated in prolate spheroidal coordinates), of the time extent left by the optical delays; the difference between the delay along the full segment and along the broken path is formed pointwise, before averaging, so that thin intervals lose no digits. The age cut depends on b and contributes a boundary term, integrated on its own quadrature. The derivative $\partial_a\partial_b$ is taken analytically, and the conformal $\hat{K}$ is computed on the same nodes and ages. The result is reported as $\hat{K}_{ab} = \partial_a\partial_b\mathbb{E}[S]/(\text{Einstein–Hilbert value } 4k^2)$.

Instruments (all within tolerance): the derivatives of the interval volume agree with an independent quadrature in spherical coordinates to 3.5×10^{-5} and 5.8×10^{-5}; a uniform optical index, a constant rescaling of space, changes no proper volume, to 9×10^{-14}; the conformal $\hat{K}$ on this grid reproduces Sec. 35 to 2×10^{-10}; a constant perturbation ($k=0$) gives zero response to 2.6×10^{-4} in units of $\hat{K}$ at the smallest k, for ages up to $T = 320\,\ell_k$; doubling every quadrature parameter changes $\hat{K}_{ab}$ by at most 3.2×10^{-5}.

Results. For each k the ages are $T = 8, 12, 16$ times $1/k$ (at least $40\,\ell_k$), and $\hat{K}_{ab}(k,T) = K_{\mathrm{loc},ab}(k) - c_{T,ab}\,k^3T$ is fitted as in Eq. (35.6):

$k\ell_k$	0.05	0.07	0.1	0.14	0.2
$K_{\mathrm{loc},ab}$ (mixed)	0.9683	0.9464	0.9045	0.8387	0.7261
K_{loc} (conformal, same ages)	0.9681	0.9441	0.8999	0.8297	0.7076
$\Delta = K_{\mathrm{loc},ab} - K_{\mathrm{loc}}$	0.0002	0.0023	0.0046	0.0091	0.0184
$c_{T,ab}$ (mixed)	0.155	0.197	0.195	0.195	0.195
c_T (conformal)	0.110	0.110	0.110	0.110	0.109

The local part is covariant. Over $k\ell_k \le 0.1$ the largest difference is $|\Delta| = 4.6\times10^{-3}$, within the criterion fixed before the run ($\le 10^{-2}$; F6.8b2, PASSED). A fit $K_{\mathrm{loc},ab} = K_{0,ab} - k^2[c_1\ln(1/k) + c_0]$ over $k\ell_k \le 0.14$ gives $K_{0,ab} = 0.9979$, i.e. a mixed stiffness

$$\mathcal{B}(k\to0) = 8K_{0,ab} = 7.983 \qquad (\text{Einstein–Hilbert } 8). \tag{36.5}$$

Two of the three coefficients of the Einstein–Hilbert quadratic form, 6 and 8, therefore emerge from the counting of intervals, and at leading order they stand in the covariant ratio.

Beyond the leading order the sectors differ. For $k\ell_k = 0.07$–0.2 the difference is a clean analytic term,

$$\Delta \simeq 0.46\,(k\ell_k)^2 \qquad (\Delta/k^2 = 0.475,\ 0.462,\ 0.462,\ 0.461), \tag{36.6}$$

and the age term of the mixed sector is 1.78 times that of the conformal one ($c_{T,ab} \simeq 0.195$ against 0.110). Neither is possible for a covariant quadratic action, local or not, which gives $\mathcal{B}/8 = \mathcal{A}/6$ at every k. The expected BD action is exactly diffeomorphism invariant, but a static field cut at age T is not a covariant configuration: the cut defines the rest frame of the field. Both differences may therefore come from the cut rather than from the action; which of the two is open. The smallest wavenumber, $k\ell_k = 0.05$, has the longest ages of the grid and departs from both regularities ($\Delta = 1.6\times10^{-4}$, $c_{T,ab} = 0.155$); it is reported and not interpreted.

The field without assuming covariance. Matter does not couple to ν. Eliminating ν from the quadratic form $\mathcal{A}\omega^2 + \mathcal{B}\omega\nu + \mathcal{C}\nu^2$ gives $\nu = -\mathcal{B}\omega/(2\mathcal{C})$, and therefore

$$\gamma(k,T) = \frac{\mathcal{B}}{2\mathcal{C}} - 1, \qquad \frac{\Phi}{\Phi_{\mathrm{GR}}} = \frac{-2}{\mathcal{A} - \mathcal{B}^2/(4\mathcal{C})}, \tag{36.7}$$

with $\mathcal{A}, \mathcal{B}, \mathcal{C}$ the coefficients of the discrete action at wavenumber k and age T (Einstein–Hilbert: 6, 8, 2, giving $\gamma = 1$ and $\Phi = \Phi_{\mathrm{GR}}$).

36.4 The optical sector

The coefficient $\mathcal{C}$ is the ν^2 sector: the light cones to second order. For $g = -\mathrm{d}t^2 + e^{2bh}\mathrm{d}\boldsymbol{x}^2$, $h = \cos kz$, two things change at second order in b.

Bent rays. The order is given by the optical distance, the minimum over paths of $\int e^{bh}\,\mathrm{d}l$ (Fermat). With $T = L + bT_1 + b^2T_2$, $T_1 = \int h\,\mathrm{d}l$ along the straight segment and

$$T_2 = \tfrac{1}{2}\int h^2\,\mathrm{d}l - \tfrac{1}{2}\iint \nabla_\perp h(l)\,G(l,l')\,\nabla_\perp h(l')\,\mathrm{d}l\,\mathrm{d}l', \tag{36.8}$$

with $G(u,u') = u_<(d-u_>)/d$ the Green function of $-\mathrm{d}^2/\mathrm{d}l^2$ with fixed ends: bending lowers the optical distance. Averaged over the phase of the wave, for a segment of length d with phase advance $p = \boldsymbol{k}\cdot\boldsymbol{e}\,d$,

$$\langle T_2\rangle = \frac{d}{4} - \frac{1}{4}\big(k^2d^2 - p^2\big)\,d\,J(p), \qquad J(x) = \frac{x^2 + 2\cos x - 2}{x^4}. \tag{36.9}$$

Interval volumes. The volume $V = \int \mathrm{d}^3z\,P(z)\,[\Delta t - T(y,z) - T(z,x)]_+$ acquires, besides the second derivatives of its integrand, a surface term $\int \delta(\ell_0)\,\ell'^2$ on the spatial projection of the diamond. Differences of nearly equal path lengths are formed analytically so that long, thin intervals lose no digits. The age cut depends on b and contributes boundary terms, among them an edge term from pairs that leave the past; it cancels a part of order T^4 of the other boundary terms exactly and is combined with it analytically.

Instruments (all within tolerance): the closed form of $\langle T_2\rangle$ agrees with rays found by direct numerical minimisation of the optical length to 8×10^{-5}; the second derivative of the volume of a test interval agrees with an independent computation (numerical roots of $\ell(b) = 0$, numerical Green function, finite differences in b) to 7×10^{-4}; a constant index gives zero to 4×10^{-15}; a constant perturbation of the whole past gives zero to 3×10^{-4} in units of $\hat{K}_{bb}$; doubling every quadrature parameter changes $\hat{K}_{bb}$ by at most 7×10^{-4}.

The age dependence is not linear. The first computation read the optical sector as the other two, by a fit linear in the age at each k, and failed by its own criterion (F6.8c, FAILED): $\hat{K}_{bb}$ falls faster than linearly with T, and the linear extrapolation carries no information. In units $\ell_k = 1$ the age-dependent part of every sector scales as $k^2F(kT)$ (the conformal and mixed data obey this to 10^{-4}), so at fixed $n = kT$

$$\hat{K}(k, n/k) = K_0 - k^2\big[c_1\ln(1/k) + c_0(n)\big], \tag{36.10}$$

and the intercept K_0 is the long-wavelength stiffness whatever F is. The revision F6.8c2 applies this reading to fresh wavenumbers ($k\ell_k = 0.06, 0.085, 0.12, 0.17$; $n = 6, 8, 10$; ages up to $167\,\ell_k$); the same reading of the conformal sector of the same rows returns $K_0 = 0.99976$ against 1.00005.

$n = kT$	6	8	10
$K_{0,bb}$ (optical)	0.99971	1.00118	0.97108
K_0 (conformal, same rows)	0.99976	0.99976	0.99977

Result. At the pre-registered $n = 8$, $K_{0,bb} = 1.00118$ (F6.8c2, PASSED), so

$$\mathcal{C}(k\to0) = 2K_{0,bb} = 2.002 \qquad \text{(Einstein–Hilbert 2)}. \tag{36.11}$$

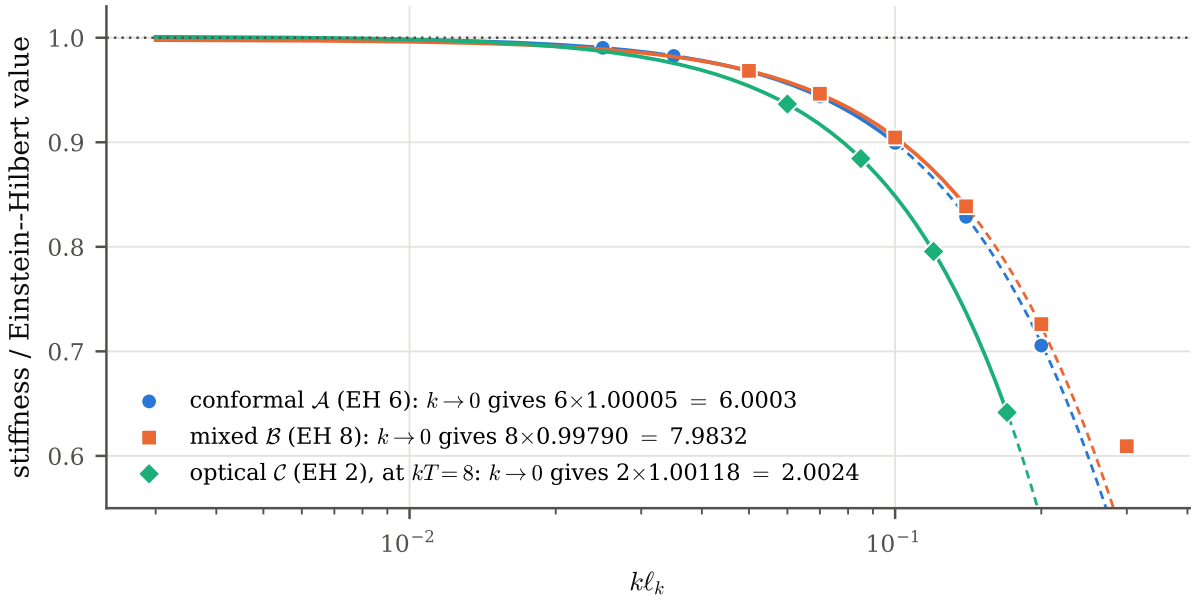


Figure 6: The three sectors of the static quadratic form approach the Einstein–Hilbert values as $k \to 0$ (F6.8a, F6.8b2, F6.8c2). Conformal $\mathcal{A}$ and mixed $\mathcal{B}$: the local part of the stiffness; optical $\mathcal{C}$: $\hat{K}_{bb}$ at the pre-registered $n = kT = 8$, which contains the age part $k^2 c_0(8)$. Lines: the fits $K_0 - k^2[c_1 \ln(1/k) + c_0]$ over the fitted range (solid) and continued (dashed). The intercepts give 6.0003, 7.983 and 2.002 against 6, 8 and 2.

All three coefficients of the Einstein–Hilbert static form, 6, 8 and 2, emerge from the counting of intervals, and with them

$$\gamma(k \to 0) = \frac{2K_{0,ab}}{K_{0,bb}} - 1 = 0.997 \text{ to } 1.000, \tag{36.12}$$

the range spanning the readings at $n = 6$ and 8 (with $K_{0,ab} = 0.99988$, the mixed sector read in the same way). $\gamma = 1$ follows from the action of the order with nothing from general relativity put in (Fig. 6). At $n = 10$ the reading is pulled by its oldest row ($k\ell_k = 0.06$, $T = 167\,\ell_k$), where doubling the quadrature also changes most; in the computation that failed, rows with $T \gtrsim 170\,\ell_k$ broke the scaling, which points to a numerical limit at large ages rather than to physics. This is not settled.

The age term of the optical sector. Within the computed range the optical sector weakens with age faster than linearly: at $k\ell_k = 0.12$ the age part $k^2 F_{bb}(n)$ grows between $n = 6$, 8, 10, 16 and 24 by increments that increase with n, whereas the conformal and mixed sectors become linear from $n \simeq 8$ ($c_T \simeq 0.110$ and 0.195). A local fit of the form $sn + qn^2$ is not stable: its coefficients depend on the range of n, and they do so for the conformal sector as well below $n \simeq 8$, where its linear asymptotics has not yet set in. The data therefore do not fix the form of the optical age term; its behaviour as $kT \to \infty$, the regime of any physical field, is open, and the predictions of the age term depend on it (Part VII; Fig. 7).

36.5 What remains

The asymptotic form of the age term in the optical sector, analytically as for the conformal one; the origin of the differences between the sectors beyond the leading order (the age cut or the action); and the field of a point mass by Fourier inversion of Eq. (36.7). These are the next computations (Part VII).

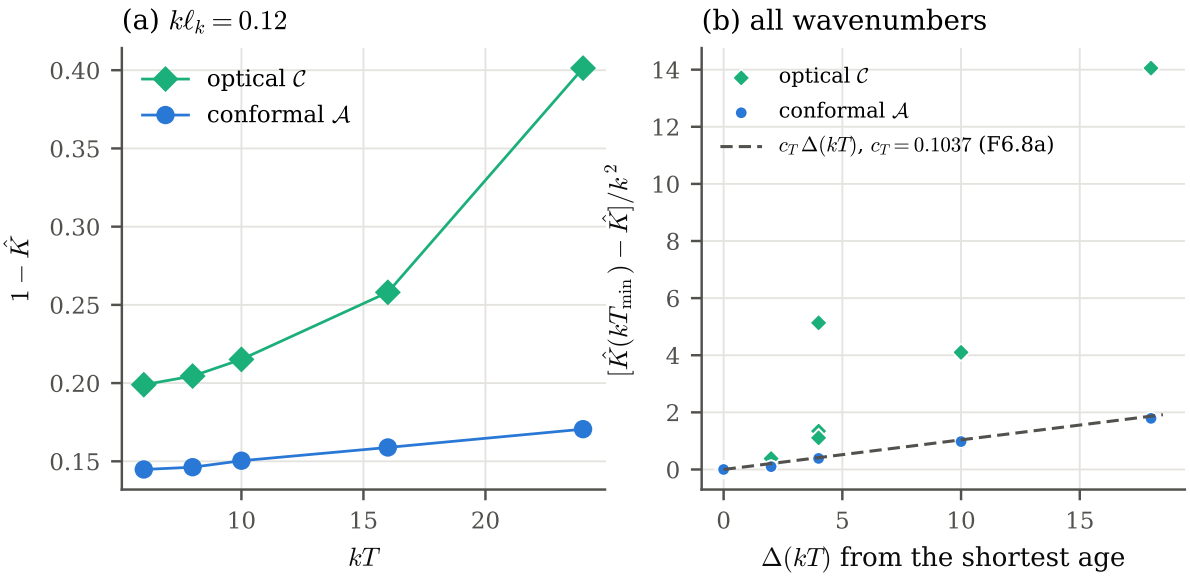


Figure 7: The age dependence of the optical sector (F6.8c2). (a) $1 - \hat{K}$ at $k\ell_k = 0.12$ for $n = kT = 6$, 8, 10, 16 and 24, optical and conformal sectors of the same rows. (b) The decrease of $\hat{K}$ from the shortest computed age, divided by k^2, for all wavenumbers: the conformal points follow the linear age term $c_T\,\Delta(kT)$ of F6.8a, the optical points lie above it and grow faster than linearly.

37 The static section to first post-Newtonian order

In the rest frame of a static source the field reduces to the clock rate Π with the metric fixed by the throughput principle,

$$\mathrm{d}s^2 = -\Pi^2 \mathrm{d}t^2 + \Pi^{-2}\mathrm{d}\boldsymbol{x}^2 = -e^{2L}\mathrm{d}t^2 + e^{-2L}\mathrm{d}\boldsymbol{x}^2, \qquad L = \ln \Pi. \tag{37.1}$$

The static field equation is the BD operator of the order acting on the logarithm of the clock rate,

$$B_g \ln \Pi = \kappa\rho_s \;\longrightarrow\; e^{2L}\big[(1-L)\Delta L + L|\nabla L|^2\big] = \kappa\rho_s \tag{37.2}$$

in the continuum limit $B_g \to \Box_g - \frac{1}{2}R$. In the weak field $\Delta L = \kappa\rho_s$, so $L = -U$ with $\kappa\rho_s \leftrightarrow 4\pi G\rho$. The choice of the logarithm is fixed by observation: for $B_g f(\Pi) = \kappa\rho_s$ the post-Newtonian parameter is $\beta = (1 - f''(1))/2$, and only $f''(1) = -1$, i.e. the logarithm in the family $(\Pi^k - 1)/k$, gives $\beta = 1$, required by the perihelion advance. The logarithm also turns the multiplicative composition of clock ratios into an additive potential.

This equation is the static section of the field to first post-Newtonian order ($\gamma = \beta = 1$). It is not the complete law: beyond this order the field is the full tensor field of Sec. 32, completed by general relativity (Sec. 38).

38 Thermodynamic completion

Beyond the linear order the coupling to the total conserved energy–momentum, including that of the field itself, determines the theory uniquely: it is general relativity [8–10]. Causal Physics adopts this completion and aims to *derive* it from the order, following Jacobson [7]: if every local causal horizon carries an entropy proportional to its area with a universal constant, the Clausius relation $\delta Q = T_{\mathrm{U}}\,\delta S$ for every local Rindler horizon, with the Unruh temperature T_{U},

implies Einstein's equations as an equation of state,

$$R_{ab} - \tfrac{1}{2}Rg_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}, \qquad G = \frac{1}{4\hbar\eta}. \tag{38.1}$$

What the order supplies:

- *Entropy proportional to area.* The number of horizon molecules measures the proper area of a horizon (K.14, Sec. 19).
- *Matter with energy–momentum.* Flows carry four-momentum and have a measured T_{ab} (Axiom 6, K.15).
- *The linear Clausius condition.* For local Rindler horizons, the Clausius relation is equivalent to $R_{ab}k^a k^b = 8\pi G\, T_{ab}k^a k^b$ for every null direction k. A clock field written in a fixed frame violates it for moving sources; the same field written in the rest frame of each source satisfies it exactly, which is linearised general relativity. The condition thus requires the field to carry the four-velocity of its sources, which is the substrate rule of Sec. 33.

What remains open is the Clausius relation beyond the linear order, with a dynamics of the order that responds to matter, and the Unruh temperature [23], which is an input from quantum field theory. The cosmological constant appears as an integration constant, consistently with its origin as a fluctuation of the number of events (Sec. 14).

Part VI
Astrophysics and the Strong Field

Astrophysics probes gravity in two directions away from the laboratory: towards strong fields (compact objects, black holes, gravitational waves) and towards weak fields over long times and large distances (galaxies, cosmology). This part states what Causal Physics says in each regime and where its own content lies.

39 The plateau is general relativity

On the plateau the order couples to matter as linearised general relativity (Part V), and beyond the linear order the coupling to the total conserved energy–momentum fixes the completion: general relativity. Causal Physics therefore inherits the astrophysics of general relativity wherever geometry is a reliable description of the order:

- compact objects and their maximum masses, determined by general relativity and the equation of state of matter;
- black holes with horizons;
- gravitational waves with two transverse polarisations, propagating at c (to 10^{-15} by GW170817 [24]), emitted with the quadrupole formula;
- the perihelion advance, light deflection, Shapiro delay and frame dragging with their general relativistic values.

These are consistency requirements, not predictions: a measured deviation from general relativity in any of them, at scales far above ℓ_k, would falsify the theory (Sec. 48).

40 Horizons in the causal picture

A black hole horizon is, in the causal picture, the limit of reconstructability (Sec. 12): the boundary beyond which an outside observer cannot keep an order-preserving description of the inside. The order itself continues: causal updates go on inside, and causal continuity is intact. [interpretation]

The entropy of a horizon is a count of the order: the number of horizon molecules is proportional to its proper area (K.14, Sec. 19). With an entropy s_0 per molecule this gives the Bekenstein–Hawking law [25, 26] and ties the discreteness length to Newton's constant, $\ell \simeq 1.075\sqrt{s_0}\,\ell_P$ (Sec. 28). [tested: K.14; bridge conditional]

What happens to the order near and inside a horizon, where the load is extreme and the geometric reading fails, is not described by the present work. Discrete corrections to horizon physics are expected at the scales ℓ and ℓ_k. [open]

41 Discreteness and non-locality in astrophysical fields

The discrete order leaves two kinds of imprint on long-lived gravitational fields.

Short scales. The static response of the order deviates from the Einstein–Hilbert form at wavelengths approaching ℓ_k: its spin-0 part weakens as $1 - K_{\rm loc} \simeq (k\ell_k)^2[4.0\ln(1/k\ell_k) + 0.9]$ and changes sign at a wavelength of $12.7\,\ell_k$ (Sec. 35). For any ℓ_k far below astrophysical scales this is invisible in astrophysics; it matters for laboratory tests of gravity at short distances (Part VII).

Long times. The non-local age term weakens the static response of a field that has existed for a time T by the relative amount $0.1\,k^3\ell_k^2 T$ (in units $c = 1$). Astrophysical fields are old, and the effect grows with their age. For the field of the Earth ($L \simeq 6400\,\text{km}$, $T \simeq 4.5\,\text{Gyr}$) a relative deviation below 10^{-2} requires $\ell_k \lesssim 1\,\text{mm}$; for the Sun at 1 AU and a Planckian ℓ_k the effect is of order 10^{-79}. What T is in a real situation (the age of the field, the age of the source, or the cosmological past of the region) is an open question of the theory; until it is settled the age term constrains ℓ_k only conditionally. The strongest conditional bounds come not from astrophysics but from short-range tests of gravity in the laboratory (Sec. 45). [derived; interpretation of T open]

Gravitational waves. The same non-locality modifies the propagation of waves with wavelengths near ℓ_k. For astrophysical waves ($\lambda \gtrsim 10^3\,\text{km}$) and any sub-millimetre ℓ_k the effect is negligible. Its form for propagating waves has not been computed. [open]

42 Galaxies and cosmology

Galaxies. Causal Physics in its present form makes no claim about a modified regime of low accelerations. Whether the discreteness or the non-locality of the order produces one, and at what scale, is open. The age term weakens the response of long-lived fields at short wavelengths; it does not strengthen gravity at large distances. In causal set theory the non-locality of the d'Alembertian has been proposed as an origin of dark matter through off-shell modes [27]; whether the order of Causal Physics, or its flows, have such modes has not been studied. [open]

The cosmological constant. The number of events in a region fluctuates by about its square root. Such fluctuations act on the large-scale geometry like a cosmological constant whose magnitude is set by the four-volume of the observable universe, and for the present universe it

is of the order of the observed value [12]; in the thermodynamic completion it is an integration constant (Sec. 38). [open]

Expansion. The growth of the order on the largest scale is the natural candidate for the expansion of the universe. A growth dynamics that produces it has not been formulated.[open]

Part VII
Predictions and Falsification

This part sets out what Causal Physics predicts, how each prediction is to be tested, and what would falsify the theory. It is the programme for the next round of tests: every entry below is formulated so that a test with a criterion fixed in advance can be written for it.

43 What counts as a prediction

A result is a *prediction* only if it is fixed before the comparison and nothing in it was calibrated on the data it is compared with. The results of the theory fall into three classes:

1. **Consistency with general relativity.** In the regime where geometry describes the order, the theory reproduces general relativity at the linear order and adopts it beyond. Agreement with the classical tests is required, not predicted.
2. **Predictions of the discreteness.** Deviations from general relativity whose *shape* is fixed by the Benincasa–Dowker kernel and whose *scale* is the single free parameter ℓ_k. These are the predictions of the theory. Data either bound ℓ_k or detect the deviation.
3. **Structural statements.** Properties of the substrate that hold independently of ℓ_k: Lorentz invariance, sharp images of distant sources, the area law of horizons.

44 Consistency with general relativity

quantity	origin in Causal Physics	status
$\gamma = 1$	throughput principle; tensor coupling in 3+1; action of the order	derived; F7.1c raw 1.000 ± 0.045; $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ emerge, $\gamma(0) = 0.997$–1.000 (F6.8a, F6.8b2, F6.8c2)
$\beta = 1$	general-relativistic completion; logarithmic static section	follows from the completion; the logarithm is chosen to match it
frame dragging	response in the rest frame of each flow	derived; F7.4a
pressure gravitates as $\rho + 3p$	coupling to T_{ab}	derived; F7.4a
quadrupole radiation, two polarisations	tensor field, GR completion	derived
equivalence principle	motion along the longest histories	derived; fall along longest chains: 2+1 passed (F5.1d), 3+1 failed (F5.1e)
Lorentz invariance	sprinkled substrate	structural

45 The window of the non-locality scale

Every prediction of the discreteness is a function of ℓ_k, and ℓ_k is not free in the whole range from the Planck length to laboratory scales.

Existing bounds. The non-locality of causal set d'Alembertians has been confronted with data for fields that propagate with the non-local operator. Comparing non-local relativistic effective field theories with collider data at 8 TeV gives $\ell_k \lesssim 10^{-19}\,\mathrm{m}$, and the ground state of an opto-mechanical oscillator gives an independent $\ell_k < 2 \times 10^{-15}\,\mathrm{m}$ [28]. In Causal Physics matter is a flow that carries its four-velocity, not a field propagated by B_ϵ, so whether these bounds apply to the scale of the gravitational sector depends on the matter sector, which is open. If the same ℓ_k governs both, $\ell_k \lesssim 10^{-19}\,\mathrm{m}$. [open]

Consequence for the local corrections. For $\ell_k \lesssim 10^{-19}\,\mathrm{m}$ the local correction of Sec. 35, of relative size $(k\ell_k)^2 \ln(1/k\ell_k)$, is below 10^{-28} at the shortest distance at which gravity is measured, $52\,\mu\mathrm{m}$ [29]: unobservable.

Consequence for the age term. The age term is not suppressed in the same way. Its relative size $\delta \simeq c_T\, k^3 \ell_k^2\, T$ grows with the age T, and with $k = 1/L$ it becomes of order one at the length

$$L_* = \left(c_T\, \ell_k^2\, T\right)^{1/3}, \qquad c_T = 0.1035. \tag{45.1}$$

Requiring $\delta < 10^{-2}$ at L gives $\ell_k \lesssim \sqrt{\delta L^3/(c_T T)}$. The outcome depends on what T is:

reading of T	T (m)	bound on ℓ_k from $L = 52\,\mu$m	L_* ($\ell_k = \ell_P$)	L_* ($\ell_k = 10^{-19}\,$m)
age of the universe	1.3×10^{26}	$1 \times 10^{-20}\,$m	$1.5 \times 10^{-15}\,$m	$51\,\mu$m
age of Earth's field	4.3×10^{25}	$2 \times 10^{-20}\,$m	$1.0 \times 10^{-15}\,$m	$35\,\mu$m
one year	9.5×10^{15}	$1 \times 10^{-15}\,$m	$6 \times 10^{-19}\,$m	21 nm
one day	2.6×10^{13}	$2 \times 10^{-14}\,$m	$9 \times 10^{-20}\,$m	3 nm

If T is cosmological, short-range tests of gravity probe ℓ_k at the level of collider bounds, and the theory says where the deviation sits: at L_*, which for $\ell_k = 10^{-19}\,\mathrm{m}$ lies at tens of micrometres, the range of current torsion balances. If T is the age of a laboratory source, the age term is out of reach. What T is (Sec. 46, P2) therefore decides whether the theory is testable at short distances now (Fig. 8). [derived; conditional on the reading of T]

Caveats. (i) The estimate uses the linear age term of the conformal sector. The optical sector weakens faster than linearly within the computed range ($kT \leq 24$, Sec. 36.4); if this persists as $kT \to \infty$, the breakdown moves to larger scales and the bounds become stronger. The asymptotic form decides the far-field law. (ii) In the mixed sector the age term is 1.78 times stronger than in the conformal one (Sec. 36.3); a covariant action cannot do this, and the difference may come from the age cut rather than from the action. (iii) At L_* the expansion in the age term breaks down, and what happens there (Sec. 46, P3) is open. The numbers above are orders of magnitude, not a verdict.

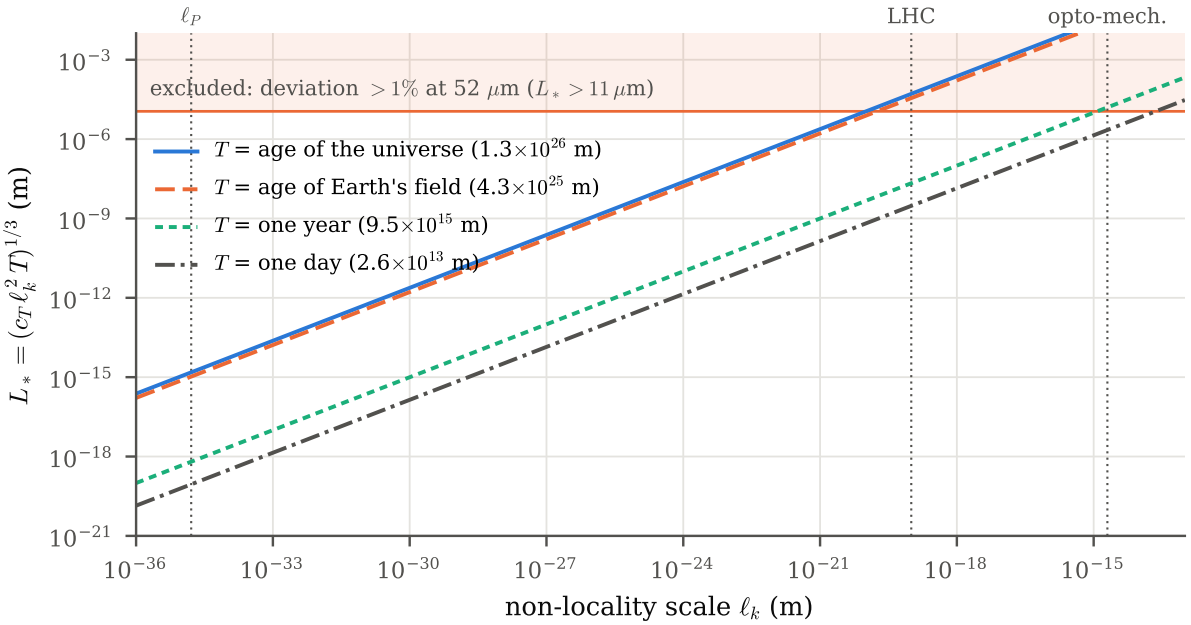


Figure 8: The window of the non-locality scale, Eq. (45.1): the length L_* at which the age term becomes of order one, against ℓ_k, for the four readings of T. Shaded: excluded by short-range tests of gravity if the deviation must stay below 1% at 52 μm ($L_* > 11\,\mu$m). Dotted: the Planck length and the bounds on ℓ_k from colliders and opto-mechanics (if the same ℓ_k governs matter). The curves use the linear age term of the conformal sector, $c_T = 0.1035$ (caveat (i)).

46 Predictions of the discreteness

46.1 P1: the static field of a point mass near ℓ_k

The static field of a point mass is fixed by the three coefficients $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ of the static quadratic form of the action of the order (Sec. 36.3), without assuming covariance:

$$\gamma(k,T) = \frac{\mathcal{B}}{2\mathcal{C}} - 1, \qquad \frac{\Phi(k)}{\Phi_{\rm GR}(k)} = \frac{-2}{\mathcal{A} - \mathcal{B}^2/(4\mathcal{C})}. \tag{46.1}$$

- *Known:* the conformal coefficient $\mathcal{A} = 6\hat{K}(k,T)$, with $K_{\rm loc} = 1 - (k\ell_k)^2[4.0\ln(1/k\ell_k) + 0.9] + \dots$ for $k\ell_k \le 0.1$ and the values of Sec. 35 beyond; the mixed coefficient $\mathcal{B} = 8\hat{K}_{ab}(k,T)$, equal to $8\hat{K}$ at leading order and larger by $\simeq 8 \times 0.46\,(k\ell_k)^2$ beyond it (Sec. 36.3); the optical coefficient $\mathcal{C} = 2\hat{K}_{bb}(k,T)$, with $\mathcal{C}(k \to 0) = 2.002$ and an age dependence that grows faster than linearly in the computed range (Sec. 36.4).
- *To compute:* the asymptotic form of the optical age term, $kT \to \infty$.
- *Prediction:* the potential $\Phi(r)$ and $\gamma(r)$ of a point mass as functions of r/ℓ_k, by Fourier inversion. The shape is fixed; only ℓ_k is free.
- *Test:* laboratory searches for deviations from the inverse-square law at sub-millimetre distances (torsion balances [29]), and light deflection and Shapiro delay at the largest k accessible. If ℓ_k obeys the existing bounds (Sec. 45), the local part of P1 is out of reach and its observable content is the age term.
- *Falsification:* if the predicted shape is excluded by the data for every ℓ_k above the discreteness length ℓ, the theory fails.

46.2 P2: the age term

A static field that has existed for a time T responds more weakly at wavenumber k by the relative amount

$$\frac{\delta\hat{K}}{\hat{K}} \simeq -0.10\,k^3\ell_k^2\,T \qquad (c=1). \tag{46.2}$$

- *Status:* derived analytically and confirmed by the full computation to 2%, in the static conformal sector (F6.8a). In the mixed sector it is 1.78 times larger, $c_{T,ab} \simeq 0.195$ (F6.8b2): the age term does not enter the sectors in covariant proportion. In the optical sector it grows faster than linearly within the computed range, $kT \leq 24$ (F6.8c2), with a form the data do not fix; it therefore shifts γ as well as the strength of the field. Its form as $kT \to \infty$ is open.
- *Open:* what T is physically (age of the field, of the source, or the cosmological past), and the form of the term in the full static propagator.
- *Consequence:* requiring a relative deviation below δ at scale L gives

  $$\ell_k \lesssim \sqrt{10\,\delta\,L^3/T}, \tag{46.3}$$

  e.g. $\ell_k \lesssim 1\,\mathrm{mm}$ from the field of the Earth at the scale of its radius ($\delta = 10^{-2}$). At the shortest tested distance, $52\,\mu\mathrm{m}$, the bound ranges from $10^{-20}\,\mathrm{m}$ (cosmological T) to $2\times10^{-14}\,\mathrm{m}$ (T of one day); Sec. 45.
- *Caveat:* the age term enters the conformal and mixed sectors in a non-covariant proportion; it may come from the way the past is cut at age T rather than from the action. A test of this is planned (F6.8d).
- *Test:* comparison of the strength of gravity of fields of different ages at the same scale, e.g. freshly assembled laboratory masses against the field of the Earth.

46.3 P3: the static pole of the spin-0 channel

The spin-0 propagator has a static pole where $\hat{K} = 0$. Without the age term it lies at a wavelength of $12.7\,\ell_k$; with it, at the length scale $L_* = (0.1\,\ell_k^2 T)^{1/3}$ of Eq. (45.1), which for T equal to the age of the universe is $\sim 10^{-15}\,\mathrm{m}$ for a Planckian ℓ_k and $\sim 50\,\mu\mathrm{m}$ for $\ell_k = 10^{-19}\,\mathrm{m}$. Whether the pole signals a standing wave, an instability of the static sector, or the limit of validity of the expansion is open. [open]

46.4 P4: the cosmological constant

The fluctuation of the number of events gives a cosmological constant of the observed order of magnitude that varies over cosmic time [12]. This is a known prediction of causal set theory that Causal Physics shares. [open]

47 Structural predictions

- **Exact Lorentz invariance of the substrate.** No preferred frame at any scale. A detected violation of Lorentz invariance would contradict Axiom 3. A Lorentz-invariant discrete substrate can still make particles diffuse slowly in momentum ("swerves" [30]); for flows that carry their four-velocity this has not been computed. [open]
- **No blurring of distant point sources by the discreteness.** Influence on the light cone passes no intermediate events, so the images of distant quasars and gamma-ray bursts are not blurred by the substrate. Observed sharp images agree with this; a detected discreteness-induced blurring would contradict it.

- **Horizon entropy proportional to proper area,** with the discreteness length tied to G through s_0 (Sec. 28).

48 Falsification

The theory is falsified, in the sense that its present form must be abandoned, if any of the following holds:

1. a deviation from general relativity is measured in the classical tests (light deflection, Shapiro delay, perihelion advance, frame dragging, binary pulsars, gravitational-wave polarisations) at scales far above any admissible ℓ_k;
2. the ν^2 sector of the discrete action gives a long-wavelength coefficient $\mathcal{C}$ different from 2: then $\gamma = \mathcal{B}/(2\mathcal{C}) - 1 \neq 1$ at large scales, $\gamma = 1$ does not follow from the action of the order, and the substrate rule is an independent postulate (checked: $\mathcal{B} \to 7.983$, $\mathcal{C} \to 2.002$, passed);
3. the static field of a point mass predicted from the three coefficients $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ is excluded by short-range gravity data for every $\ell_k \geq \ell$;
4. a violation of Lorentz invariance of the substrate is detected;
5. the images of distant point sources are found blurred in a way attributable to the discreteness of spacetime.

49 The next tests

test	question	needs
F6.8e	the asymptotic age term of the optical sector, $kT \to \infty$ (analytic, as for c_T)	F6.8c2
F6.8d	the differences between the sectors beyond leading order: from the age cut or from the action?	F6.8b2
F8.1	$\Phi(r)$, $\gamma(r)$ of a point mass in units of ℓ_k; comparison with short-range data	F6.8e
F8.8	the physical meaning of the age T; fields of different ages	theory of T
F6.9	the age term in the full static propagator	F6.8c

Each of these will be registered with its criterion before it is run, and its outcome reported whatever it is.

Part VIII
Discussion and Conclusion

50 What has been shown

Starting from events and their order, Causal Physics reaches the following, each step tested or derived:

1. The substrate is a Lorentz-invariant causal set, and its geometry can be read blindly from order and number.
2. Matter is a flow that carries its four-velocity; localisation requires this memory, and the memory is inertia. The flows have a conserved energy–momentum tensor.

3. The order responds to matter, and the way it responds fixes γ: a fixed order gives $\gamma = -1$, narrowing cones give $\gamma = 1/d_s$, conservation of throughput gives $\gamma = 2/(d_s - 1)$, which is 1 in three spatial dimensions.
4. In three spatial dimensions the throughput principle is the coupling of linearised general relativity to T_{ab}, and a substrate rule realising it makes pressure and motion gravitate and drags the cones.
5. That rule is not an independent postulate: with the proper time of a flow counted by the order, it is the linear response of an order whose action has the Einstein–Hilbert stiffness.
6. The discrete Benincasa–Dowker action has that stiffness: 6.0003 against 6 in the static conformal sector, 7.983 against 8 in the mixed sector and 2.002 against 2 in the optical sector, computed with nothing from general relativity put in; $\gamma = 1$ follows from it. The discreteness adds a non-analytic correction and non-local terms that grow with the age of the field, and beyond the leading order the sectors differ.

51 Limitations

- **Expected values, not dynamics.** The action of the order is evaluated as an expectation over sprinklings of a given geometry. That a dynamics of the order selects such causal sets, the manifold-like phase, is assumed, as in causal set theory generally.
- **Finite ages.** The coefficients are computed for ages up to a few hundred ℓ_k; the age term of the optical sector grows faster than linearly there, and its asymptotic form is not known. Beyond the leading order the sectors are not in covariant proportion; whether this comes from the age cut or from the action is open.
- **The substrate rule as an input in the tests.** The tests of the coupling (F7.1c, F7.4a) build the order according to the rule. They show what the rule implies, not that nature uses it; the derivation from the action supplies the reason.
- **Matter as a self-bound object.** Flows are localised by their memory of direction, but a particle that holds itself together without an external source has not been obtained.
- **The quantum.** The theory is classical. Superposition and measurement have an interpretation, not a dynamics.
- **The age term.** Its coefficient is established; its physical reading, what T is, is not, and it may depend on how the past is cut. The reading decides whether short-range gravity tests probe ℓ_k now (Sec. 45).
- **The scale ℓ_k.** Existing bounds on non-local fields give $\ell_k \lesssim 10^{-19}\,\mathrm{m}$. Whether they apply to the gravitational sector depends on the matter sector, which is open.
- **Finite numerics.** Laboratory results come from finite regions and finite numbers of sprinklings, with the limits stated in each test.

52 Outlook

The road from here to predictions is short and concrete: the asymptotic age term of the optical sector, then the static field of a point mass from the three coefficients, compared with short-range gravity data, and the physical meaning of the age T. In parallel, the thermodynamic route is to turn the completion by general relativity from an adopted principle into a derived one, and the matter sector is to find the self-bound flow.

53 Conclusion

Gravity in Causal Physics is the response of a causal order to the matter that flows through it. The order slows its clocks, narrows its cones and adds events so that its throughput is kept, and in three spatial dimensions this is exactly the coupling of general relativity. The reason is the action of the order itself: counting the intervals of a causal set gives the Einstein–Hilbert stiffness, and the counting also says where it stops being exact. That is where the predictions of the theory are.

Appendices

A The Causality Lab

A.1 Purpose and architecture

The Causality Lab [1] is an open Python code base written to test Causal Physics step by step. It has three layers:

- a *core*: sprinkling of causal sets in 1+1 to 3+1 dimensions, causal relations and links, interval counts, the discrete and expected Benincasa–Dowker operator and action (flat and curved), transport of flows with capacity and carried four-velocity, and blind instruments that read geometry from the order;
- *tests*, one file per test, each with a header that states the question, method, prediction, null model, criterion, level of conclusion, the instruments it depends on and an estimate of its cost;
- a *runner* that executes a test (`python run.py <ID>`), records the full parameters, the seed, the version and the output, and refuses to change the criteria of a full run.

Deterministic computations (expected values by quadrature) carry their own convergence checks; stochastic ones (sprinklings) are replicated with independent seeds before a verdict is final.

A.2 Rules

The methodological principles M1–M7 (Sec. 29) are enforced by the runner and the test headers; a revision of a test states its reason in its header. Calibrations (phase K) establish instruments and carry no physical conclusion.

A.3 Tests cited in this paper

test	question and result	verdict
K.1	geometry readable from the order in 3+1 from $N \approx 3 \times 10^4$	calibration
K.2	causal front measures the Shapiro delay (0.98 of GR) and deflection (0.84–0.87)	calibration
K.3	clock rate readable from the density of chains	calibration, replicated
K.11, K.11c	curved BD operator tends to $\Box_g - \frac{1}{2}R$ (conformal geometry, near and far field)	calibration
K.13a	BD operator in a geometry with bent light converges, density and cone parts separately	calibration
K.14	horizon molecules measure the proper area (exponent 1.96 ± 0.06 in 3+1)	calibration
K.15	energy–momentum tensor of flows; four-momentum conserved exactly by $F = \lvert P \rvert$	calibration, replicated
K.16b, K.16c	blind instrument reads $\gamma = -1, 0, 1$ (2+1) and $-1, \frac{1}{3}, 1$ (3+1)	calibration
F0.3	degree of an event grows with N: no bound K_0	mechanism
F1.1	Gauss's law for a steady source, $J/S = 1.003$–1.004	mechanism
F1.3	far field of a source: Green function with Gauss strength	mechanism
F1.4	superposition fails at second order as predicted	mechanism, passed
F1.8	capacity limits the flow, not the number of relations	mechanism (seed-sensitive)
F2.2, F2.4, F2.5	maximal chains follow the geodesics of $\Pi^2\eta$ (2+1, 3+1)	mechanism, passed
F3.6	a flow with carried direction is localised and moves with its source	mechanism, passed, replicated
F4.1	flows that meet head-on stick: merged fraction 0.82 (2+1), 0.77 (3+1), merged flow at rest, no bounce	mechanism, passed
F5.1	free fall with a direction carried in flat coordinates, or keeping to the longest chains from the origin: no fall	mechanism, instrument failed; failed
F5.1c	longest-chain history started at rest: inward, $R = 0.34 \pm 0.05$	mechanism, failed
F5.1d	longest-chain history started at the apex of a thrown ball falls, $R = 1.069 \pm 0.051$ (2+1)	mechanism, passed
F5.1e	the same in 3+1 does not fall, $R = -0.13 \pm 0.12$; longest chains read the geometry, $R_{\max} = 0.85$	mechanism, failed
F6.1, F6.2	local growth with bounded degree: no manifold, no 3D space	mechanism
F6.4a	2D orders with BD action: manifold phase and transition reproduced; height of the crystalline phase not	calibration, failed
F6.8a	EH stiffness of the expected BD action, 6.0003; correction and age term	chain, passed
F6.8b2	mixed stiffness, 7.983 against 8; covariant at leading order, sectors differ beyond (revision of F6.8b, whose instruments failed)	chain, passed
F6.8c	optical sector read linearly in the age: the age dependence is not linear	chain, failed
F6.8c2	optical sector read at fixed kT on new wavenumbers: $\mathcal{C}(0) = 2.002$ against 2	chain, passed
F7.1a	cones narrowed without added events give $\gamma \approx \frac{1}{2}$ (2+1), as derived	mechanism, failed
F7.1b	throughput principle in 2+1: $\gamma \approx 1.85$ against 2	mechanism, undecided
F7.1c	throughput principle in 3+1: $\gamma = 1.000 \pm 0.045$, 0.972 ± 0.074	mechanism, undecided
F7.4a	tensor coupling of counter-streaming flows, $R = 2.401 \pm 0.053$ against 2.361	mechanism, passed

A verdict answers the question the test asked. FAILED for F7.1a means that cones which only narrow do not give $\gamma = 1$, which was the derived expectation; UNDECIDED for F7.1b and F7.1c means that the statistical comparison with the control metric was too weak for a formal verdict, while the raw values agree with the prediction.

A.4 Availability

The code, the test headers and the recorded outputs of every run quoted here are available at https://github.com/pavelsikula92/Causal-Lab-. Every number in this paper can be reproduced with `python run.py <ID>`.

B Relation to the first version

The first published version of this programme [31] stated it in a broad form. This appendix relates the two: what is kept, what is reformulated, and what is no longer claimed. Every change results from a test or a derivation in the present work.

B.1 Kept

- The primacy of causality over geometry; events without intrinsic properties; the causal order as the deepest structure.
- Causal continuity: influence is conserved at every event.
- Finite capacity, now as a bound on the flow through an event.
- The local response variables Π_v, I_v and $Z_v = I_v/\Pi_v$, and the Causal Resistance Functional as the action of a flow.
- The picture of c as the maximal congestion-free throughput and of c_{eff} falling where the substrate is loaded. It is now the throughput principle, which gives $\gamma = 1$ in three spatial dimensions.
- The separation of the causal, bridge and observational layers.
- The ontological themes: observation and causal resolution, branching and merging, horizons as limits of reconstructability, superposition as coexisting continuations.

B.2 Reformulated

- **The substrate.** A graph with bounded degree K_0, locality by graph distance and time as a count of updates has a rest frame. It is replaced by a Lorentz-invariant causal set, as required by the theorem of Bombelli, Henson and Sorkin and supported by the tests F0.3, F1.8, F6.1 and F6.2.
- **Dimension.** The effective dimension is no longer defined by the growth of graph balls; it is read from the order (fraction of related pairs, blind instruments).
- **The dynamical principle.** The Causal Resistance Functional is the action of a flow; the dynamics of the order is the Benincasa–Dowker action.
- **Gravity.** The inverse-square law from flux through spherical shells is replaced by the response of the order: clocks, cones and density, with the tensor coupling derived and tested.
- **Special relativity.** The Lorentz factor is no longer derived from clocks at rest relative to the substrate; Lorentz invariance is a property of the substrate.

- **Gravity changes the causal order.** Near matter the cones narrow and events are added; causality itself (no loops, conservation of flow) is untouched.

B.3 No longer claimed

- The cosmological anchor $a_0 = cH_0/2\pi$ and the low-acceleration regime built on it, including the reinterpretation of dark matter. The anchor is not derived from the present formulation, and its mechanism relied on the bounded-degree substrate.
- The plateau $D_{\text{eff}} \approx 3$ as an attractor of a bounded-degree dynamics.
- Threshold effects in strong gravity (suppression of the redshift of white dwarfs, environment-dependent delays of gravitational waves), and strong-field properties of a scalar field equation. Beyond the linear order the theory is completed by general relativity.
- The projection of time and length through the Hubble scale and the projection of impedance through the vacuum electromagnetic impedance.

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Competing interests

The author declares no competing interests.

Data and code availability

The code of the Causality Lab, the headers of all tests with their pre-registered criteria, and the recorded outputs of every run quoted in this paper are available at `https://github.com/pavelsikula92/Causal-Lab-`. The version used for this paper is archived as [TODO: release tag and Zenodo DOI of the code].

Licence

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